

DCSS328

Deep Learning Application

Orientation

Lecture 0

Hanseob Kim

Fall 2026

Korea University

Computer Science and Software Engineering



Immersive Media eXperience



IMX Lab Advertisement



VR/AR/MR and Metaverse

Human-Computer Interaction

UI/UX and Universal Design

AI-Generated Digital Human

Psychological
Affective Computing

VR-based Digital Therapy

VR/AR/MR and
Metaverse

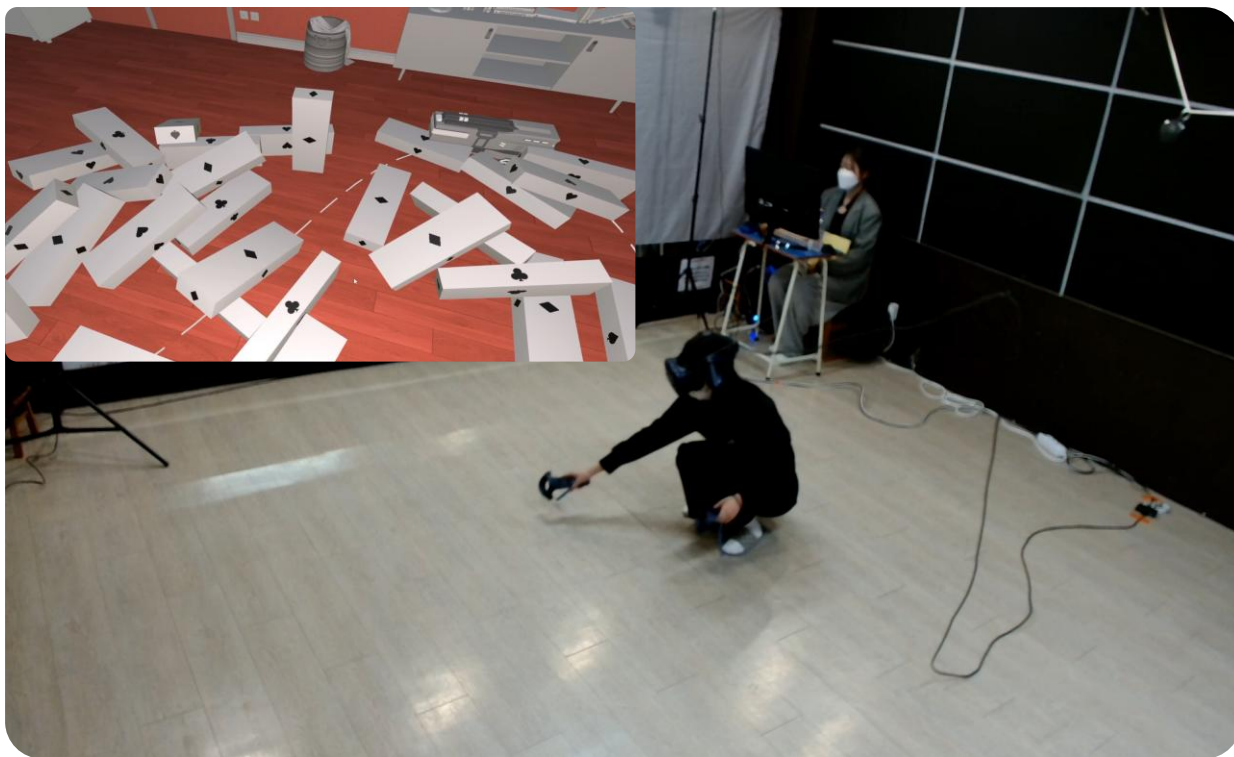
Human-Computer
Interaction

UI/UX and
Universal Design

인공지능 기반
디지털 휴먼

인지·심리 기반
감성 컴퓨팅

가상현실 기반
디지털 치료



VR/AR/MR and
Metaverse

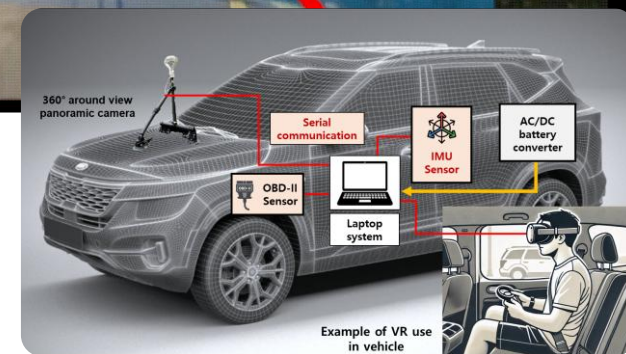
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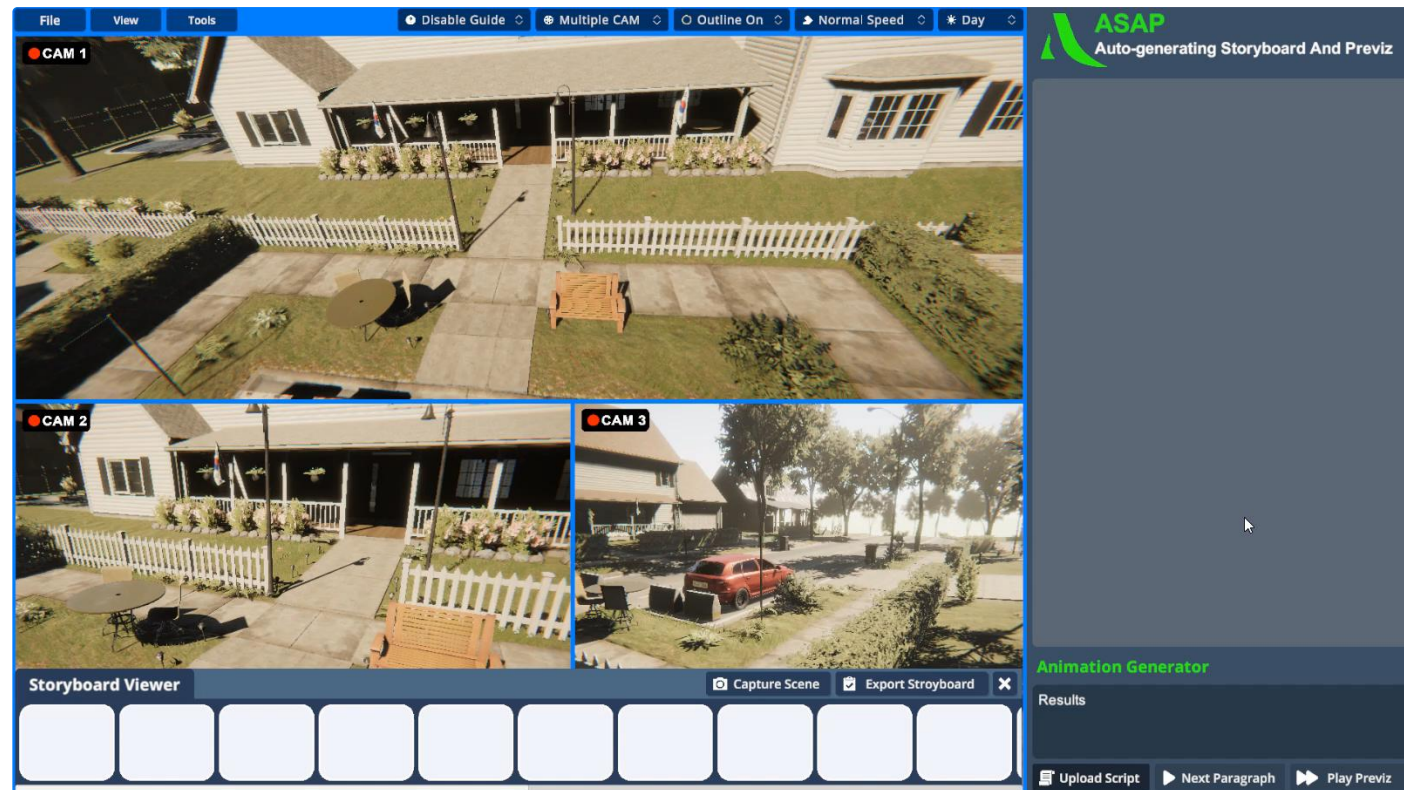
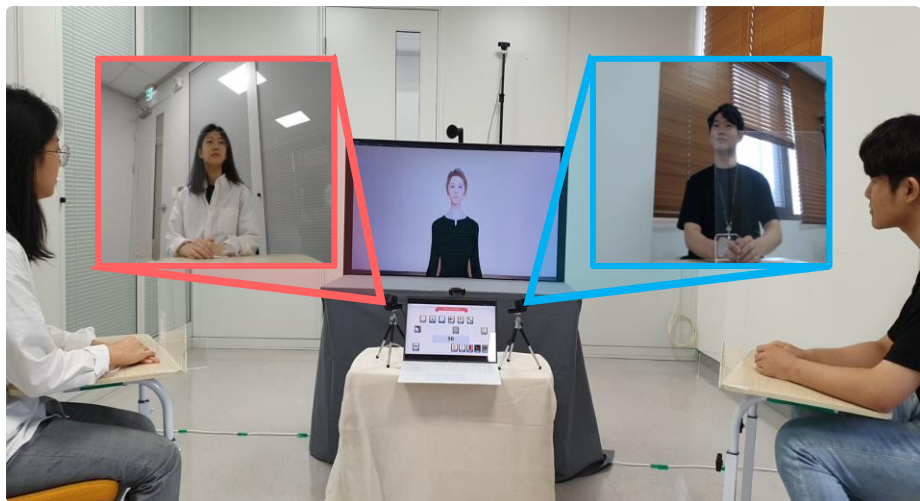
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VR/AR/MR and
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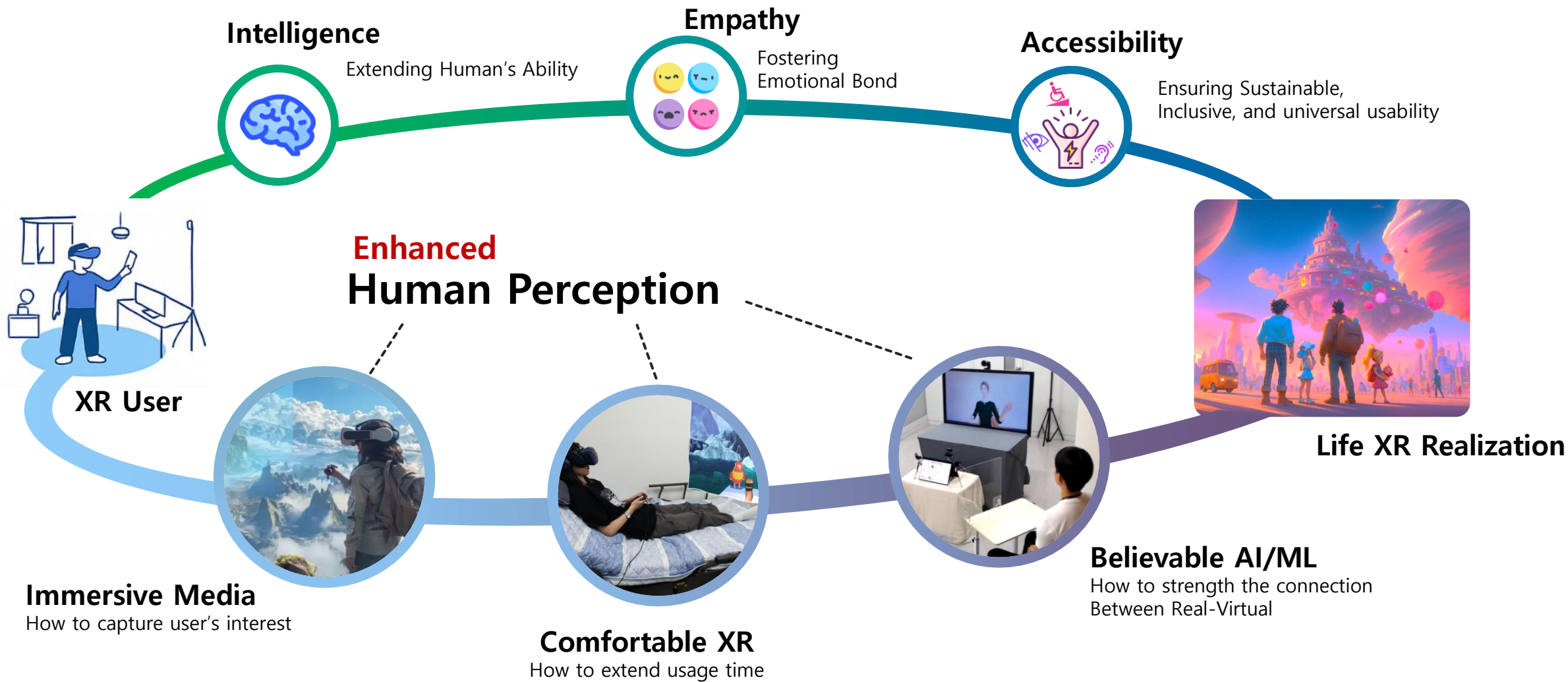
UI/UX and
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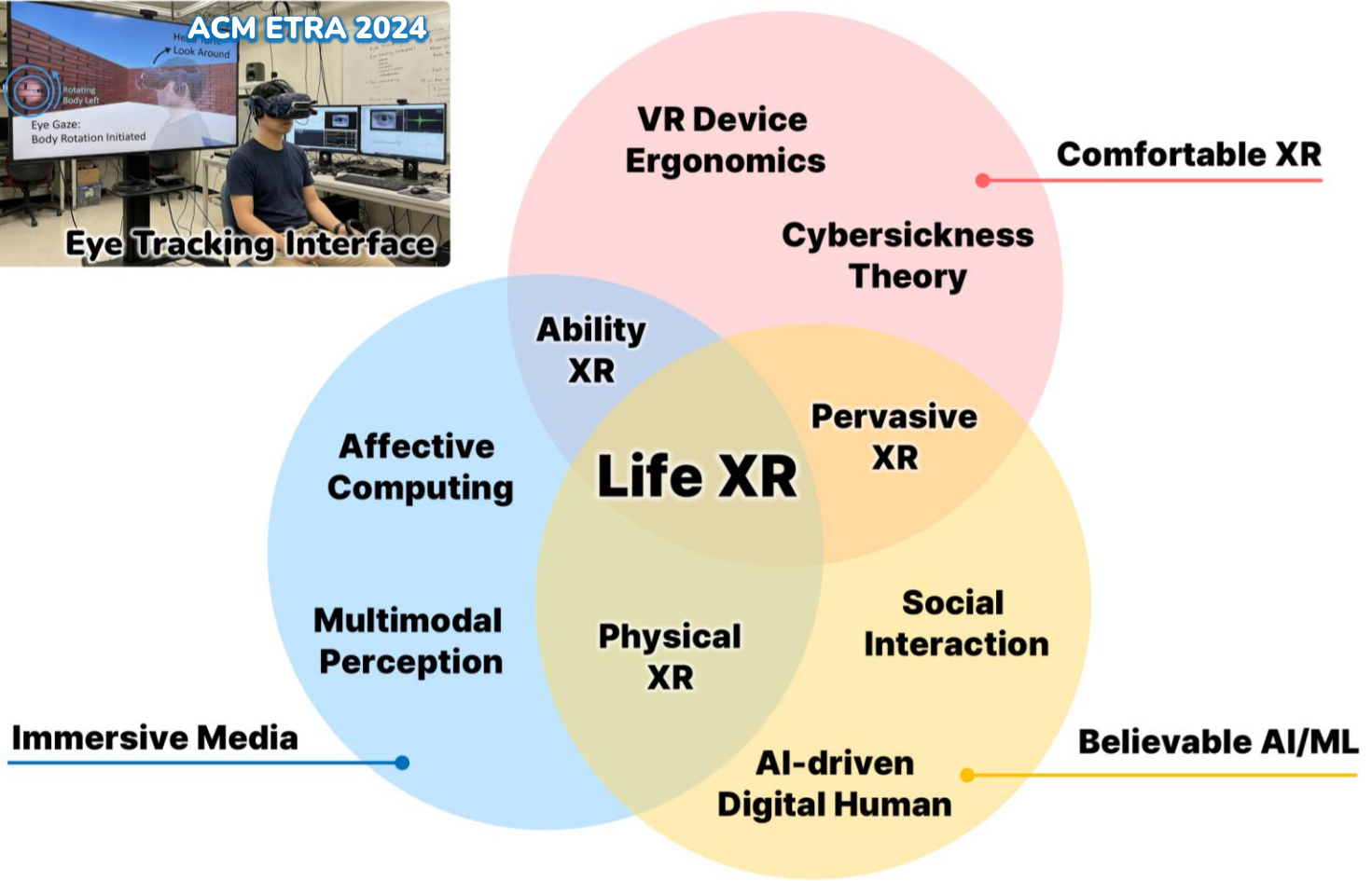
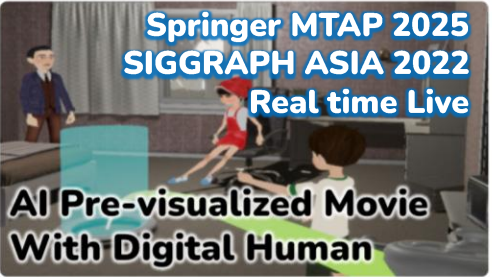
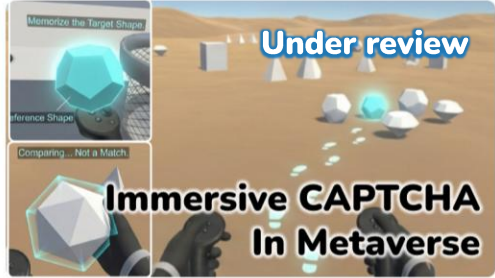
AI-Generated Digital
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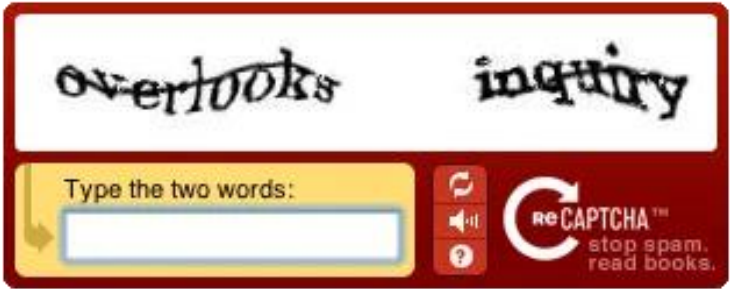
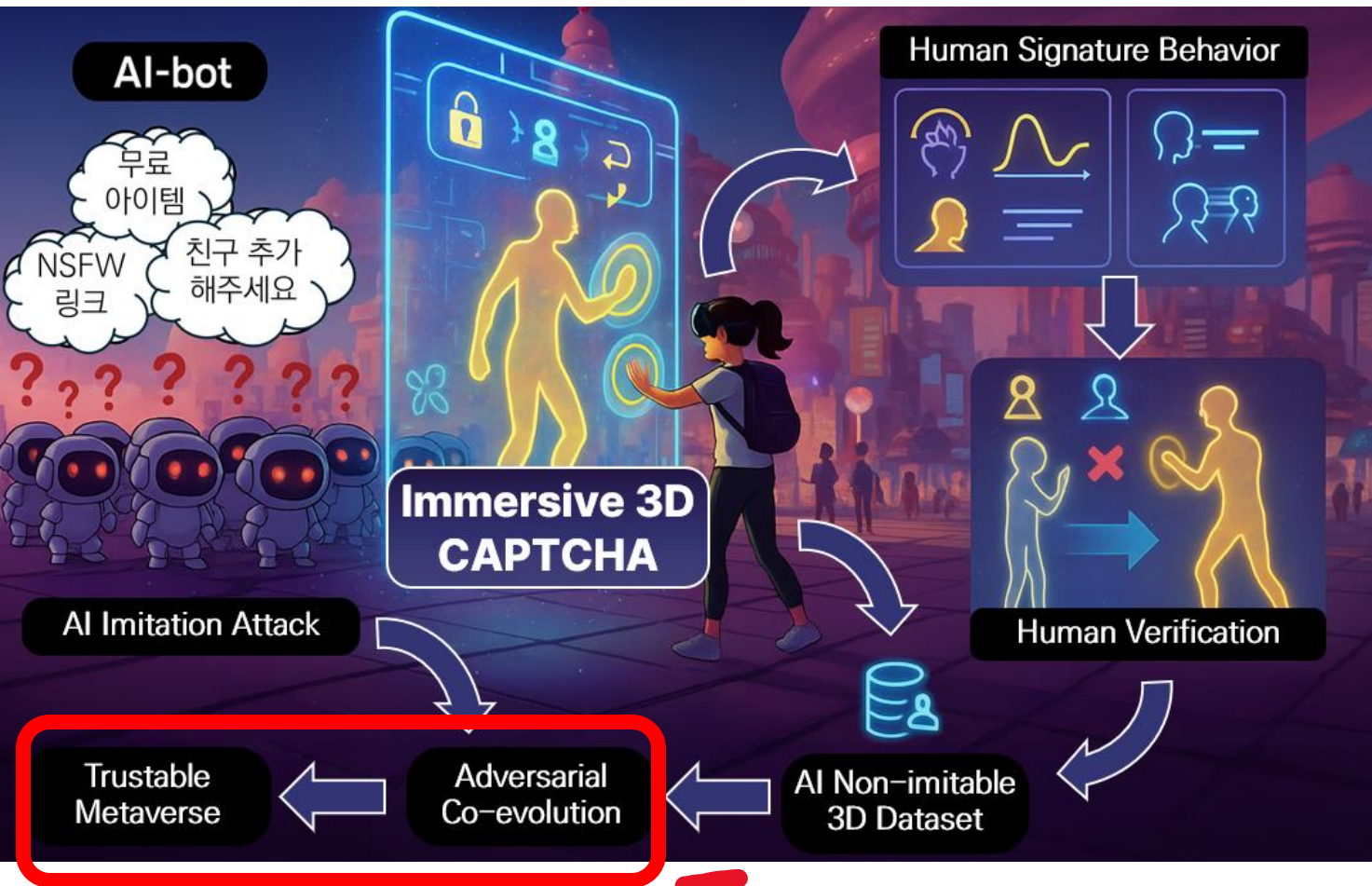




Latest funding project from Korea



How to improve the security level of the metaverse?



Answer: Use Human's 3D Data to make CAPTCHA

Blue Team
Propose Immersive 3D CAPTCHA

VS

Red Team
Verify Robustness Against AI Attacks

I'm looking for self-motivated students who are interested in VR/AR/XR and HCI/UX

**If you want to join our lab
please contact me**

khseob0715@korea.ac.kr

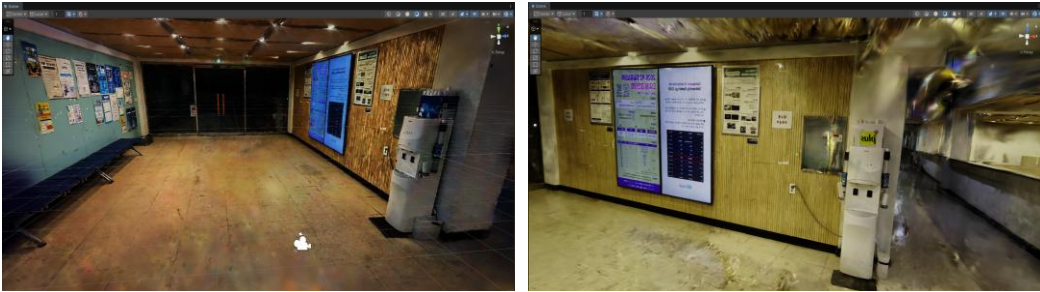
Course Road Map

Week	Course Plan	Key Questions
1	Recent trends in deep learning models (LLM, Agentic AI, World Model, etc.)	What we will learn in this course and develop for the final-term project
2	Machine Learning Recap	How to learn from data, and improve efficiency?
3	Deep Learning Recap	How can we train a model using only a small amount of data?
4	3D Space Reconstruction 1: 3D Gaussian Splatting	How can a 2D photo be transformed into a 3D space?
5	3D Space Reconstruction 2: NeRF	How can a neural network generate a scene from a viewpoint it has never seen?
6	3D Human Modeling 1: Diffusion-based 3D model	How can a 2D photo be transformed into a 3D human model?
7	3D Human Modeling 2: RigNet	How can a 3D human model be animated?
8	Mid-term Exam	Assessment
9	RL Animation 1: DeepMimic	How can an AI agent learn to control a physically simulated character?
10	RL Animation 2: Adversarial imitation	How can a character learn natural movements by imitating motion-capture data?
11	Speech AI 1: ASR & TTS	How can AI recognize and generate human speech?
12	Speech AI 2: Voice Cloning	How can AI clone and adapt a person's voice using only a few samples?
13	LLM-RAG-Reasoning Knowledge 1	How can we choose between prompting, RAG, and fine-tuning for a specific task?
14	LLM-RAG-Reasoning Knowledge 2	How can an LLM retrieve information, use external tools, and act as an intelligent agent?
15	Term Project Presentation	What memory did you reconstruct ?
16	Final Exam	Assessment

Course Road Map

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14	LLM-RAG-Reasoning Knowledge 2
15	Term Project Presentation
16	Final Exam

- 3D Virtual Environment
- 3D Virtual Human Body
- Animating Virtual Human
- Speaking Virtual Human
- Thinking Virtual Human



A. Video Taken in the Past



B. 3D Mesh Generated from Video



C. Walking Back into Past Space in Virtual Reality



MakeAGIF.com

Course Road Map

- ❖ **Life Logging** Saving of real-life as a photo, and then, Re-generating it to the 3D virtual environment

Can AI transform our recorded memories into interactive experiences?

You !



Our Final Goal ! XR Memory Application

Course Road Map

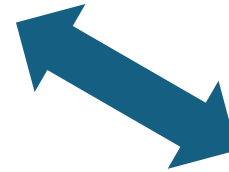
- ❖ **Life Logging** Saving of real-life as a photo, and then, Re-generating it to the 3D virtual environment

Copy your body and use
as **virtual human**

With AI-generated
speech and body gesture



Interactive conversation based on
your information using LLM, RAG



Reconstruct your memorable scene as virtual environment

Materials

Lectures will be based on slides uploaded to the LMS, primarily developed from ***Research Paper***

- **After the onsite lecture, the pre-recorded video of the lecture will be released on LMS**

Most of the materials needed in the course will be released on LMS

Evaluation

Item	Ratio	Item	Ratio
Attendance	5%	Assignments	50%
Midterms	20%	Final Exam	20%
<i>Occasional assessments</i>	%	<i>Attitude</i>	5%
<i>Engagement</i>	%	<i>etc</i>	%
Total	100%	Score visibility	Open(V) Closed()
Evaluation Method Description	- Attendance will account for 5% of the final grade. 0.5 points will be deducted for each absence. - The midterm and final exams will account for 20% and 20%, respectively. - Assignments will account for 50%, consisting of four short assignments worth 5% each (20% total) and a term project worth 30%. - For the term project, students will integrate multiple AI models into a single working XR-based interactive application/system. - Overall class participation and learning attitude will account for 5%.		
Attendance Policy	No grade may be assigned for absences of 1/3 or more of the class days, except in cases of unavoidable absences deemed necessary by the instructor.		

Assignment/Team Presentation

Assignment, report, and project names	How to Write	Submission Deadlines	Submission Types and Methods
A1: 3D Space recon	Use Scaniverse Tool	9 Week	TBA (LMS)
A2: 3D Human recon	TBA	9 Week	TBA (LMS)
A3: RL Animation	TBA	16 Week	TBA (LMS)
A4: Voice clone	TBA	16 Week	TBA (LMS)
Term project/presentation	Project/PPT	15 Week	Github/In-class Presentation

DCSS328

Deep Learning Application

Recent Trends in Deep Learning

What we will learn in this course

Lecture 1

Hanseob Kim

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Today's Core Question

“Why can AI solve an Olympiad Problems, but still struggle to clean a Room”



- Relatively simple rule
- Clear Answer



- Noise
- Physics rule
- Long planning & Latency
- Safety

NVIDIA GTC Taipei 2026 Keynote

<https://www.youtube.com/watch?v=wSp6AiNIrsY>

2012 ALEXNET



PERCEPTION AI

SPEECH RECOGNITION
DEEP RECSYS
MEDICAL IMAGING

GENERATIVE AI

DIGITAL MARKETING
CONTENT CREATION

AGENTIC AI

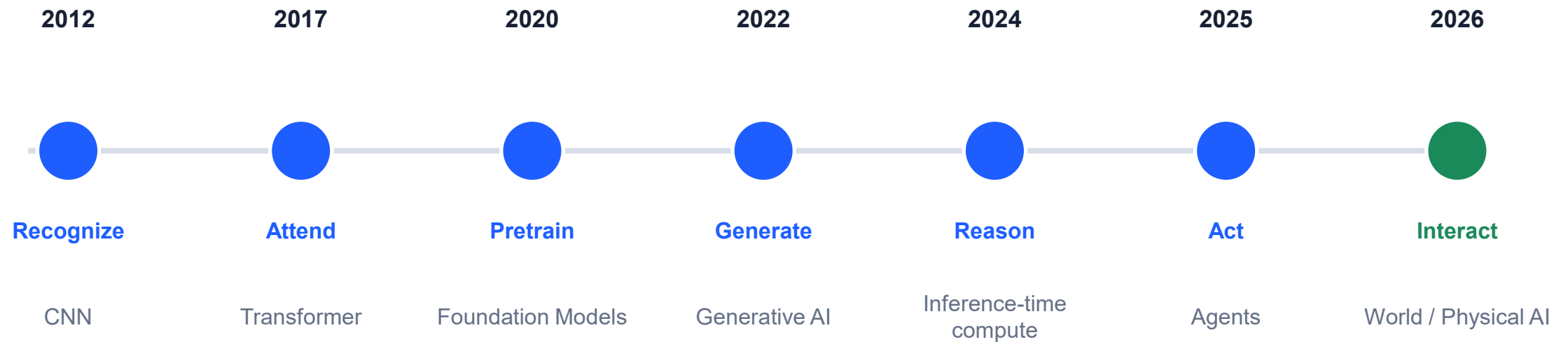
CODING ASSISTANT
CUSTOMER SERVICE
PATIENT CARE

PHYSICAL AI

AUTONOMOUS VEHICLES
GENERAL ROBOTICS

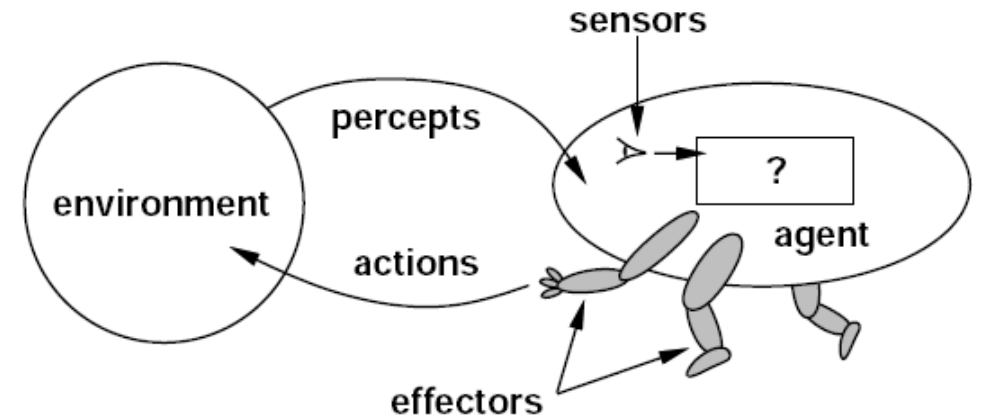


The AI Trends and its Capability Change



Model capability **expanded from pattern recognition to systems that reason, use tools, and interact with environments.**

The AI Trends and its Capability Change



Key five words

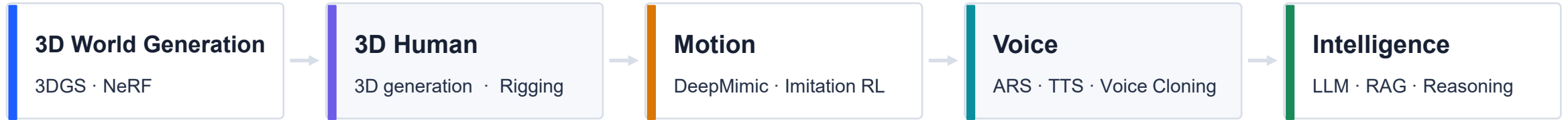


However, Prediction is not disappeared; still underneath everything

The difference is that modern systems wrap prediction with reasoning, tools, memory, simulation, and feedback.

From AI trends to our course

Our course focuses on the components needed to build an interactive AI experience in XR



TERM PROJECT

Integrate multiple AI models into one working XR-based interactive application/system.

01

Foundation Models & Efficient AI

From “train one model per task” to “adapt and compose pretrained capabilities.”

Deep Learning Application in 2026

The Old Pattern vs. the New Pattern

TRADITIONAL DEEP LEARNING

Dataset A → train Model A → Task A

Dataset B → train Model B → Task B

Dataset C → train Model C → Task C

Strong specialization, repeated training cost.

MODERN FOUNDATION MODEL

Large-scale pretraining



General representations



Prompting / retrieval / **Fine-Tuning**



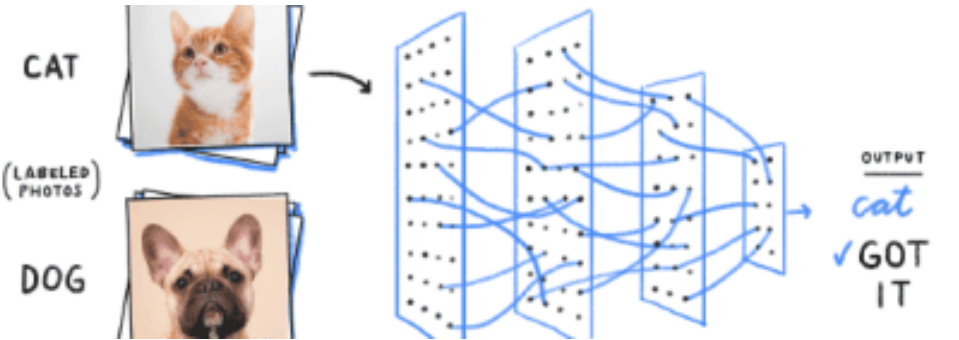
Many downstream tasks

The model becomes a **reusable** capability layer

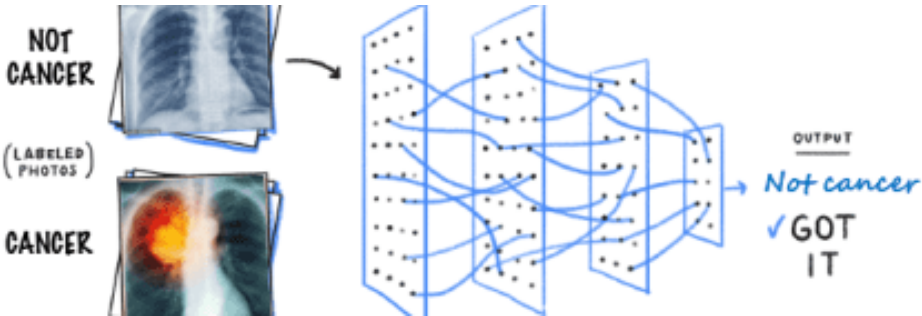
.

The Old Pattern vs. the New Pattern

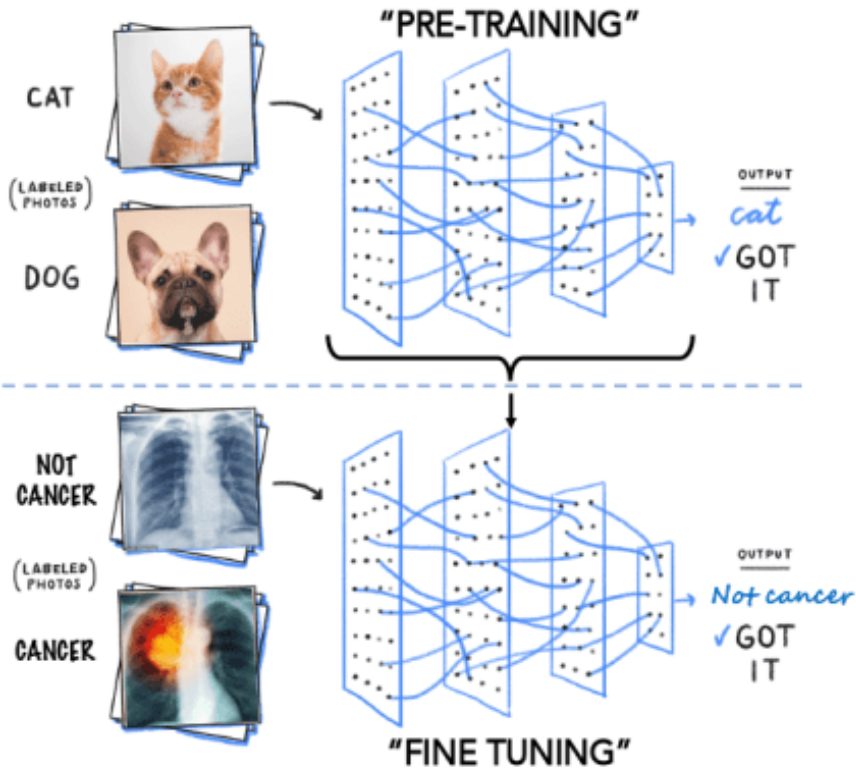
TRADITIONAL DEEP LEARNING



Strong specialization, repeated training cost



MODERN FOUNDATION MODEL

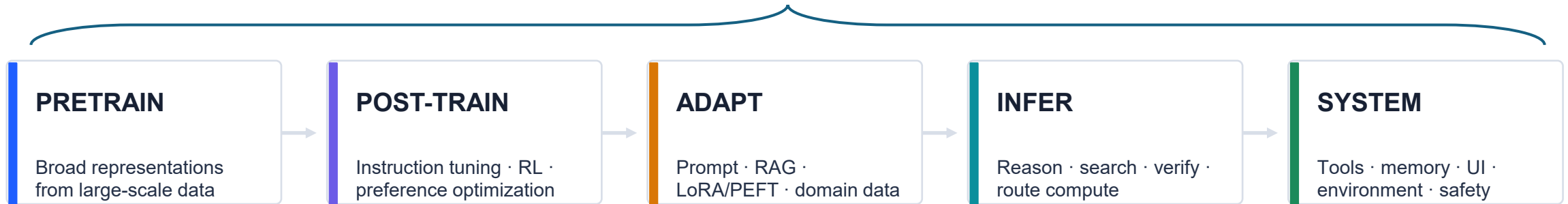


reusable trained weights

AI Modern AI Stack

Progress happens at multiple layers — not only in model architecture.

AI Model can be organized into Five Layers



Modern AI engineering often means selecting where to **inject task-specific information and where to spend compute.**

In the past the focus was on how well the model itself was trained,

Now the important engineering decision is where to allocate resources among pre-training, post-training, RAG/context, inference-time compute, and agent/system.

Scaling Has More Than One Axis

Intelligence gains increasingly depend on how computation is allocated

Past

More Large Model + More Large Data + More Time for Train → GOOD AI

Model Scale

Parameters
architecture
Mixture-of-Experts

Makes the brain
itself bigger

Data Scale

Quantity
quality
deduplication
synthetic data

Trains with
more/better materials

Training Compute

Pretraining
post-training
reinforcement learning

Trains for longer
and better

Inference Compute

Sampling
search
verification
tool use

You think longer when
you receive a problem.

A model does not have to spend the same amount of computation on every query.

Scaling Has More Than One Axis

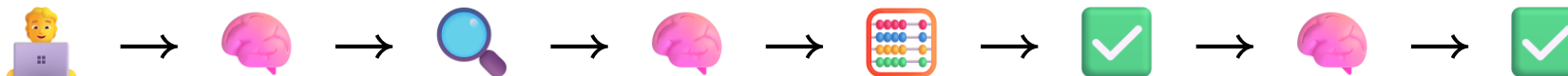
Intelligence gains increasingly depend on how computation is allocated

Easy Question



Low compute / Shorter Inference

Hard Question



High compute / Longer Inference

When Capability Converges, System Trade-offs Matter

Public leaderboards move quickly; production decisions increasingly depend on more than benchmark score.

QUALITY

Does it solve the task?

COST

How much compute per request?

LATENCY

Can it respond in time?

RELIABILITY

Does it succeed repeatedly?

TOOL USE

Can it operate in a workflow?

“Best model” is becoming a task- and system-dependent choice.

Model A

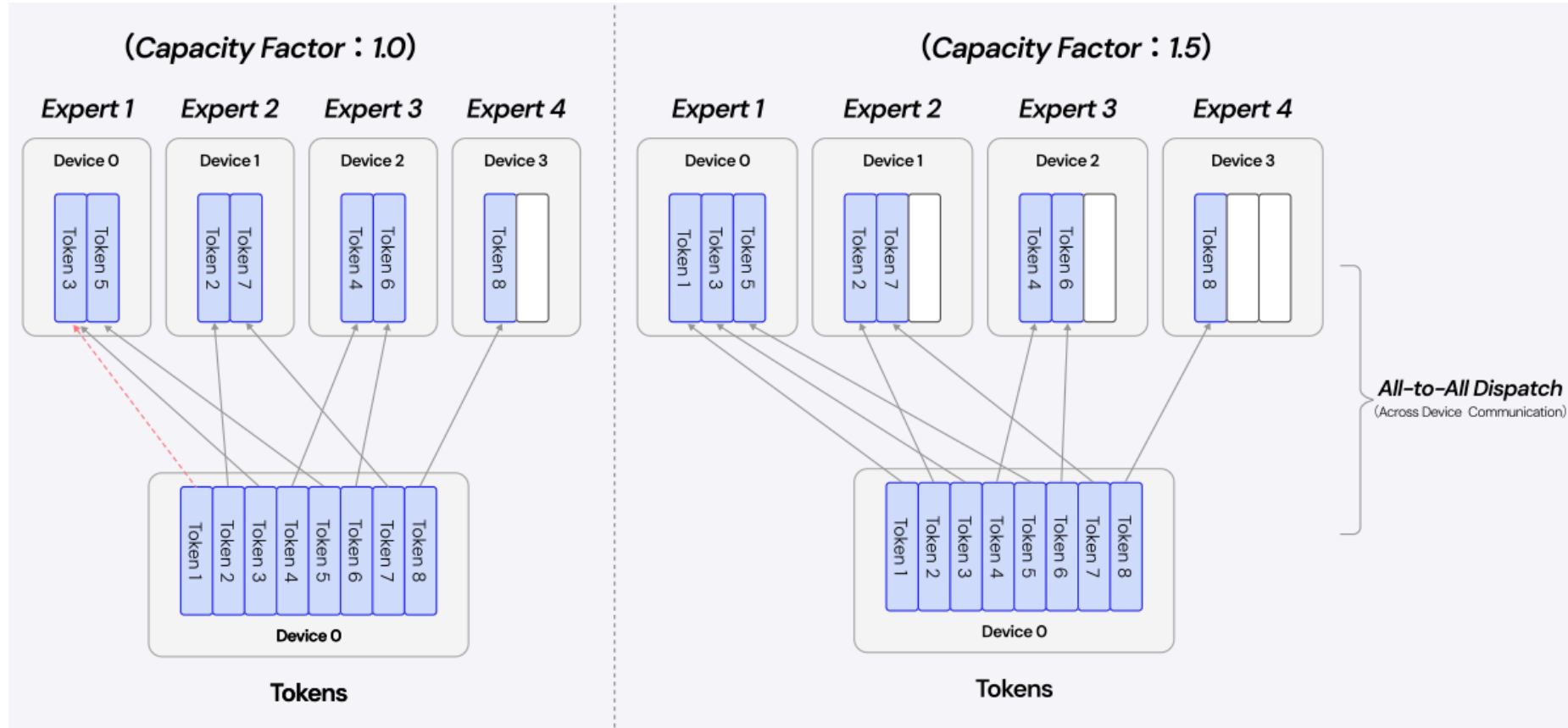
Accuracy : 95 %
Response Time : 20 s
Fee : \$0.2
Usability : Good

vs

Model B

Accuracy : 93 %
Response Time : 1 s
Fee : \$0.01
Usability : So So

Sparse Models and Mixture-of-Expert



Why it matters

- Higher capacity at lower active compute
 - Useful for cloud, edge, XR, and robotics constraints
 - Better memory / throughput trade-offs
- Total parameters $\neq$ active parameters**

The Adaptation Toolbox

Where should task-specific information enter the system?

PROMPTING

Change instructions,
not weights

RAG

Add external knowledge
at inference

LoRA / PEFT

Update a small fraction of
parameters

QUANTIZATION

Lower precision to reduce
memory / compute

DISTILLATION

Transfer behavior to a smaller
model

ROUTING

Choose the right model /
compute budget

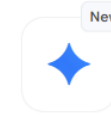
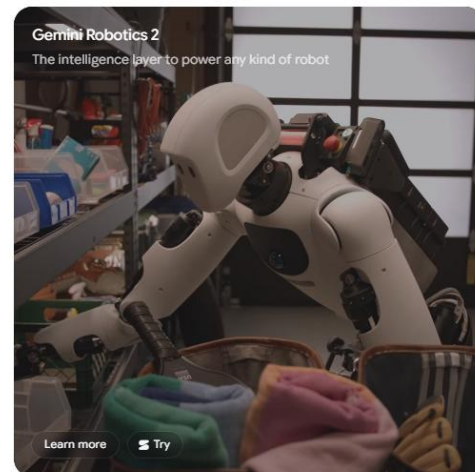
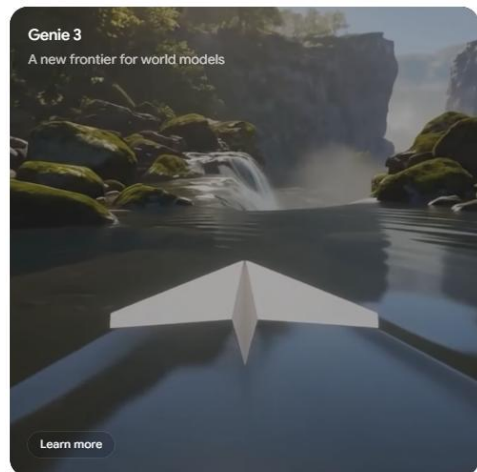
Preview of Week 13: prompting vs. RAG vs. fine-tuning is a design decision, not a hierarchy.

Recent Trends

Native Multimodal Intelligence

Google DeepMind · Current Gemini / Omni model family (2026)

Text, image, audio, video, code, tools
inside one reasoning and action layer



Gemini

Frontier intelligence with action

TRY IT IN

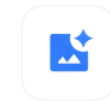
[Gemini app](#) >

[Google AI Studio](#) >

[Google Antigravity](#) >

[Gemini Enterprise Agent Platform](#) >

[Learn more](#)



Nano Banana 🍌

Create and edit images with Gemini Image

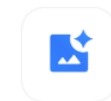
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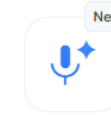
Imagen

Our leading text-to-image model

TRY IT IN

[Google AI Studio](#) >

[Gemini Enterprise Agent Platform](#) >



Gemini Audio

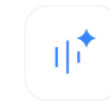
Advanced real-time audio models, built on Gemini

TRY IT IN

[Google AI Studio](#) >

[Gemini Enterprise Agent Platform](#) >

[Learn more](#)



Lyria

Our most advanced music generation model

TRY IT IN

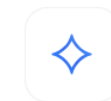
[Gemini app](#) >

[Google Flow Music](#) >

[Google AI Studio](#) >

[Gemini Enterprise Agent Platform](#) >

[Learn more](#)



Gemma

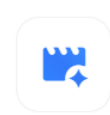
Our most capable open models

BUILD WITH GEMMA

[Google AI Studio](#) >

[Run Gemma](#) >

[Developer docs](#) >



Gemini Omni

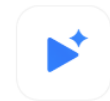
Create anything from anything, starting with video

TRY IT IN

[Gemini app](#) >

[Google Flow](#) >

[Learn more](#)



Vevo

Our leading video generation model

TRY IT IN

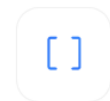
[Gemini app](#) >

[Google Flow](#) >

[Google AI Studio](#) >

[Gemini Enterprise Agent Platform](#) >

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Gemini Embedding

State-of-the-art multimodal embedding model

BUILD WITH GEMINI EMBEDDING

[Gemini API docs](#) >

[Google Cloud docs](#) >

<https://deepmind.google/models/>

02

Reasoning & Test-Time Scaling

The model does not have to spend the same amount of computation on every problem.

Deep Learning Application in 2026

Why Reasoning Became a Separate Capability

Think of reasoning as computation allocation over time — not as human-like consciousness.

FAST RESPONSE

Question



One forward generation



Answer

Great for routine tasks, but limited opportunity to explore alternatives or verify mistakes

DELIBERATE RESPONSE

Question



Propose → test → revise → verify



Answer

Spend extra inference compute when difficulty or uncertainty is high.

Train-Time Scaling vs. Test-Time Scaling

TRAIN-TIME

More pretraining / RL / post-training



A better policy before deployment

TEST-TIME

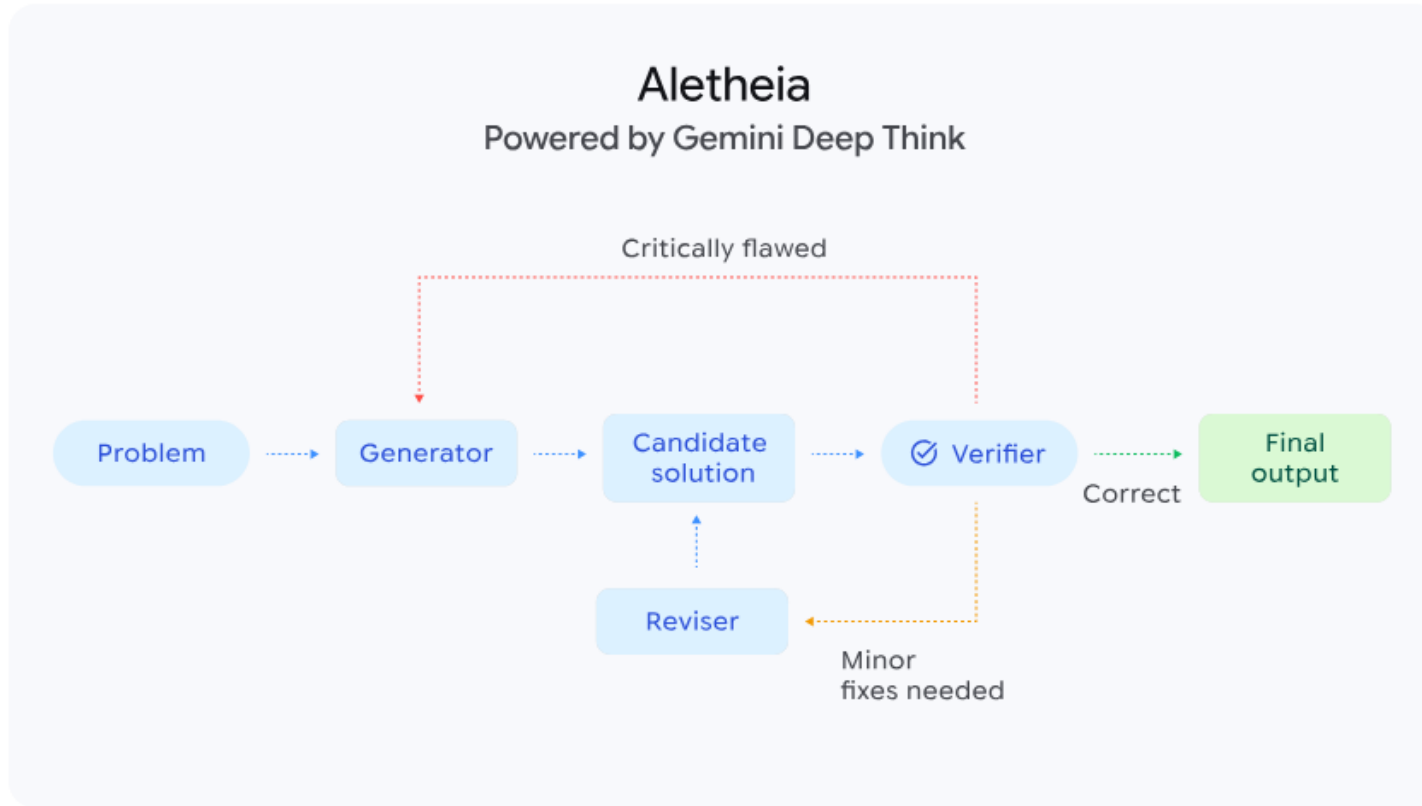
More search / sampling / verification / tool use



Potentially a better answer for this query

How much computation should this specific query receive?

Reasoning Can Be a System Loop



Overview of Aletheia, a math research agent powered by Deep Think that can iteratively generate, verify, and revise for research-level math problems.

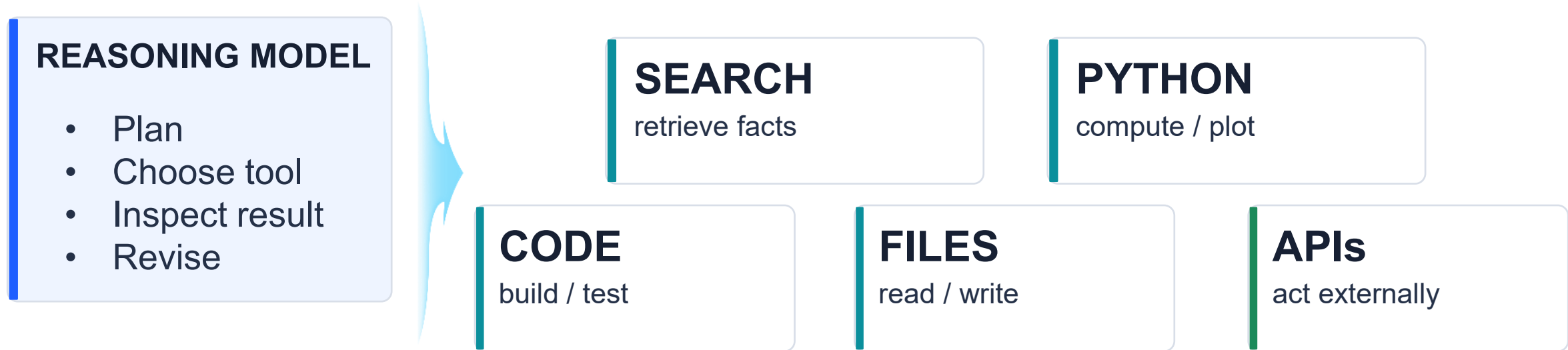
<https://deepmind.google/blog/accelerating-mathematical-and-scientific-discovery-with-gemini-deep-think/>

The “reasoner” is increasingly an orchestration of generation, checking, tools, and selective retries

Reasoning + Tools

Example loop: search → compute → inspect → revise → answer

A model **can pivot based on external evidence** instead of guessing through missing information.



The important capability is knowing when to use which tool and **then incorporating the tool result** into the next reasoning step

Benchmarks Are Moving — and Breaking

State of the art is conditional on the benchmark and **Evaluation Protocol.**

HARDER BENCHMARKS

Frontier performance on Humanity's Last Exam improved by roughly 30 percentage points in one year.

BENCHMARK SATURATION

Once an evaluation saturates, it stops telling us where systems fail.

BENCHMARK QUALITY

A review highlighted invalid -question rates from 2% to 42% across widely used evaluations.

SOTA can not defined by one metric

Jagged Intelligence

The same system can be extraordinary on one task and surprisingly weak on another.

VERY HARD

Gemini Deep Think achieved gold-medal-level performance at the 2025 International Mathematical Olympiad.

SEEMINGLY SIMPLE

On ClockBench, the top model read analog clocks correctly 50.6% of the time vs. 90.1% for humans.

Do not infer general competence from a single spectacular demo.

Limits of Reasoning

- 1 Wrong verification** A confident checker can validate a flawed path.
- 2 Excess compute** More tokens can add latency / cost without better quality.
- 3 Tool error** A correct plan can fail because browsing, code, or APIs fail.
- 4 Benchmark issues** Leakage, ambiguity, and invalid questions distort comparisons.
- 5 Monitorability** Internal reasoning traces are useful signals, but safety and control remain active research topics.

03

Multimodal & Generative AI

Text is becoming one modality among many, e.g., audio, image, video, code, 3D data, action

And generation is moving toward dynamic, interactive worlds.

Deep Learning Application in 2026

From Modality Pipelines to Native Multimodality

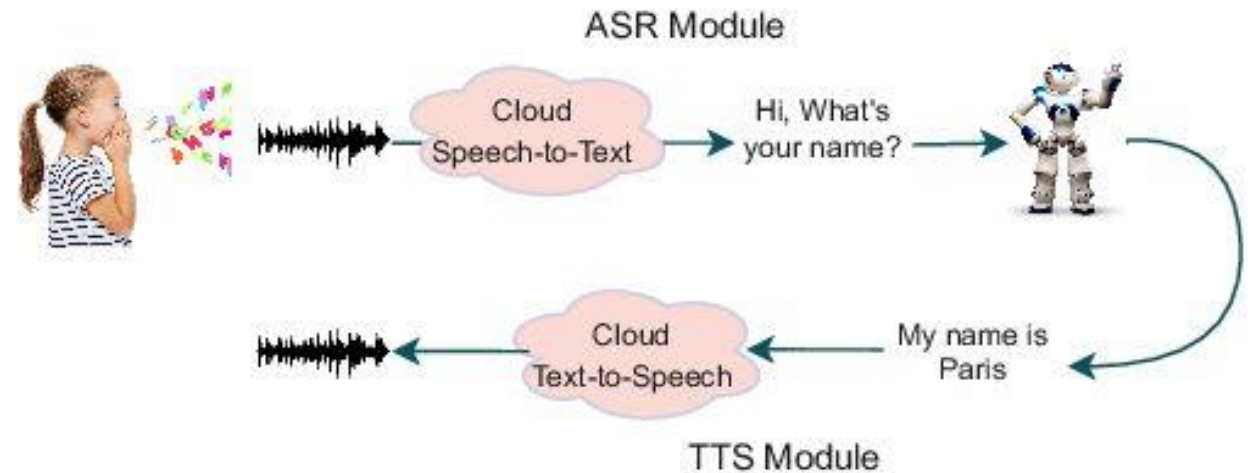
In past, Separate Module used

Image encoder → text model

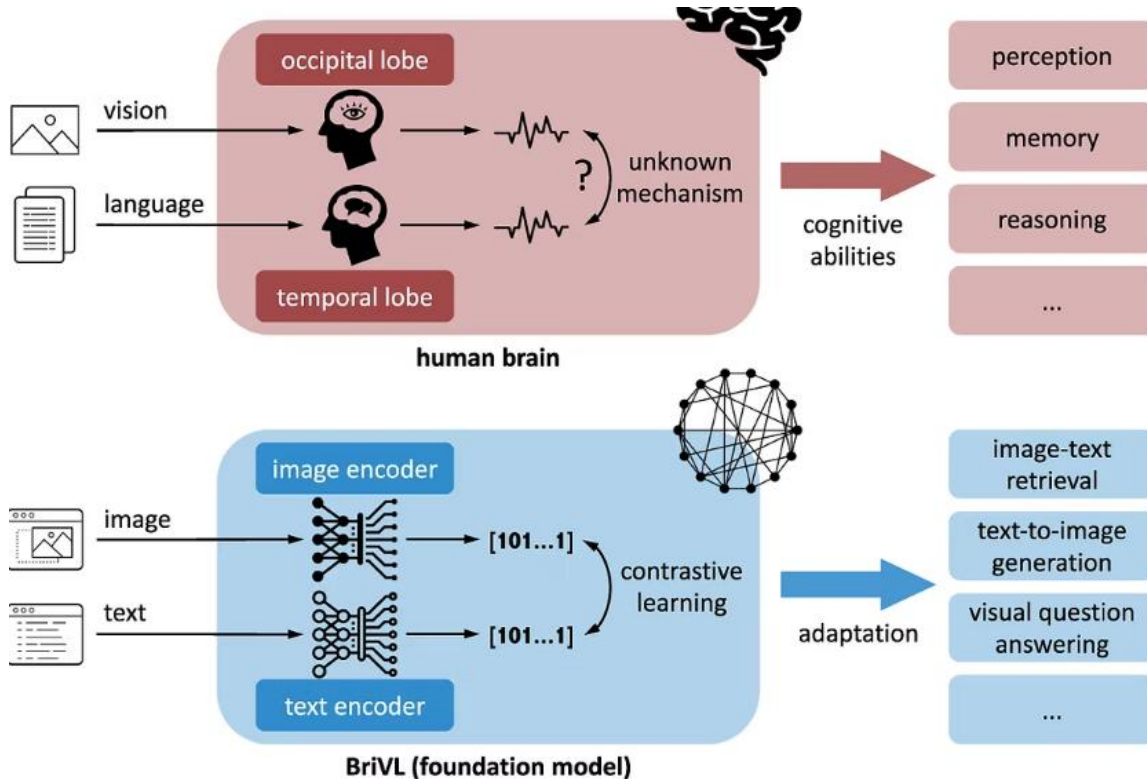
ASR → LLM → TTS

Video model → captioner

Each modality often had a specialized interface.



From Modality Pipelines to Native Multimodality



NATIVE MULTIMODAL ERA

Text + image + audio + video + code



Shared reasoning / tool-use layer



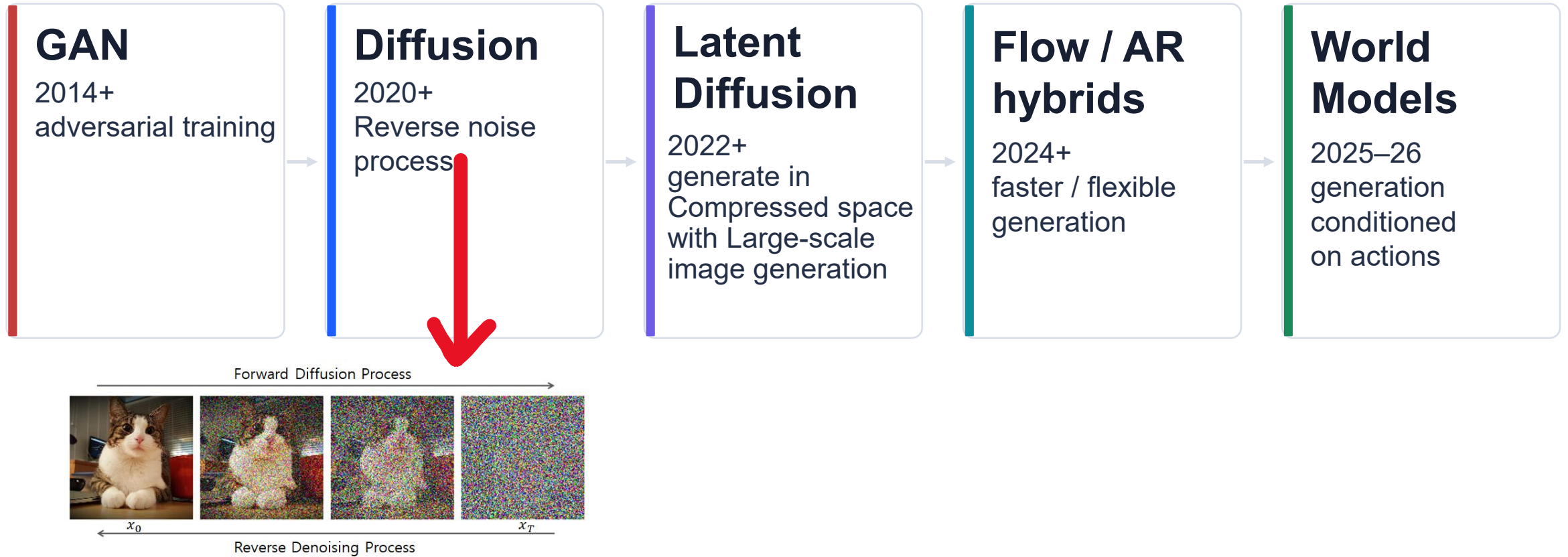
Multimodal output or action

Modality switching becomes part of one interaction loop.

The important advantage is that the **system does not always need to change everything to text first.**

History Generative Modeling

<https://www.youtube.com/watch?v=IxHXQpk5kkg>



Trajectory: static samples → coherent/consistent sequences → action-conditioned environments

Video Generation Is Becoming a World-Learning Problem

Evaluation increasingly asks whether generated video reflects world behavior — not only whether it looks real.

VISUAL FIDELITY

- Sharp Images
- temporal consistency
- cinematic quality

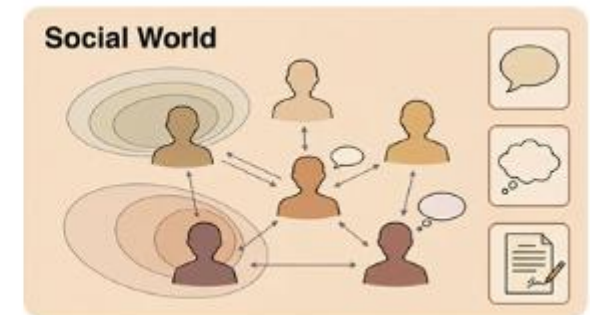
SEMANTIC CONTROL

Text / image steering
object control
camera / event control

PHYSICAL BEHAVIOR

Buoyancy
collision-like interactions
navigation / persistence

Visual realism and world correctness are not the same thing



Will AI understand Dynamics in physical World and Social World??

<https://deepmind.google/models/veo/>

Promptable Perception: SAM(segment anything model)

A specialist foundation model can become a perception tool inside a larger agentic system.

TEXT

"striped red umbrella"

POINT / BOX

interactive visual prompt

EXEMPLAR

show one example object

<https://www.aidemos.meta.com/segment-anything>



Isolate any part of an image or video instantly

Just type a phrase, and the model instantly finds every matching object across your media.



Turn your segments into dynamic, editable media

Experiment with new possibilities through one-click effects like blurring, cloning, and more.



Create reusable templates that speed you up

Advanced media tools for all. Save, download, and share with the community.

SAM 3 / 3.1

Detect → segment → track

2026 DIRECTION

- Unified detection + segmentation + tracking
- Text / exemplar / visual prompts
- Public playground
- Specialist vision tools become callable components.

04

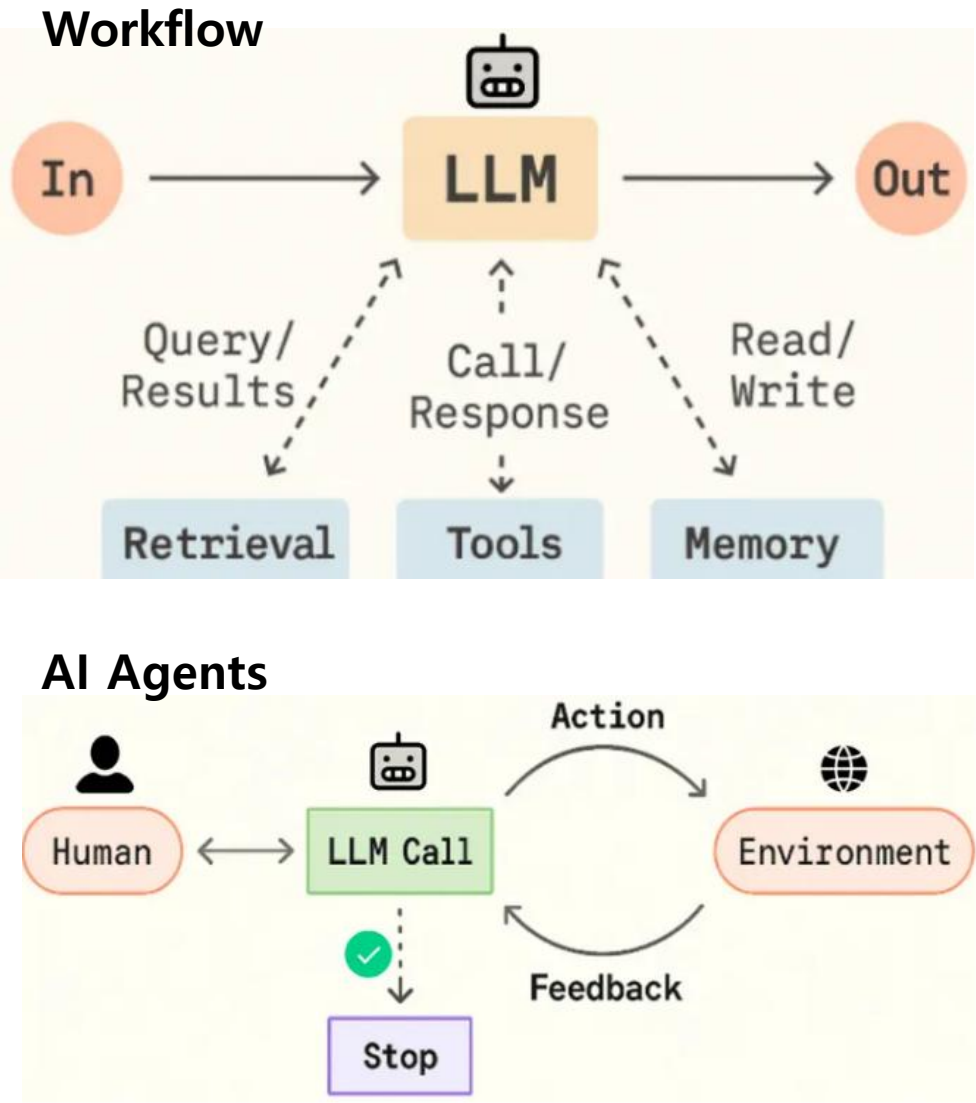
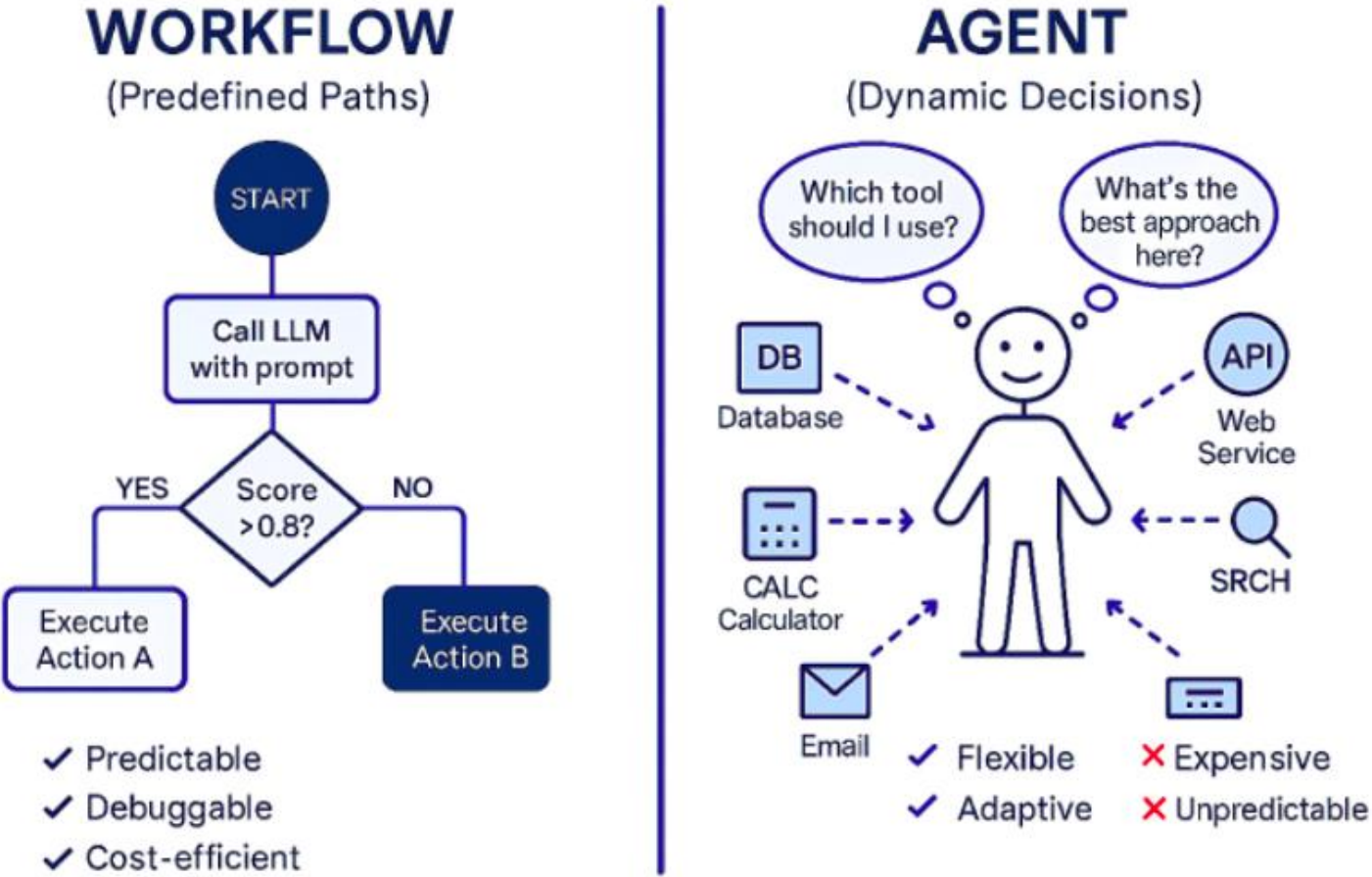
Agentic AI

When a model can select tools, observe results, and keep going, it becomes part of an acting system.

Output is no longer just text; Output may change an external environment!

Deep Learning Application in 2026

Workflow vs. Agent



<https://ai.plainenglish.io/ai-workflows-vs-ai-agents-vs-multi-agentic-systems-a-comprehensive-guide-f945d5e2e991>

A Minimal/Basic Agent Loop

Learn / Reflect

Improve performance
based on **feedback**

Sense / Observe

Collect inputs from
the environment

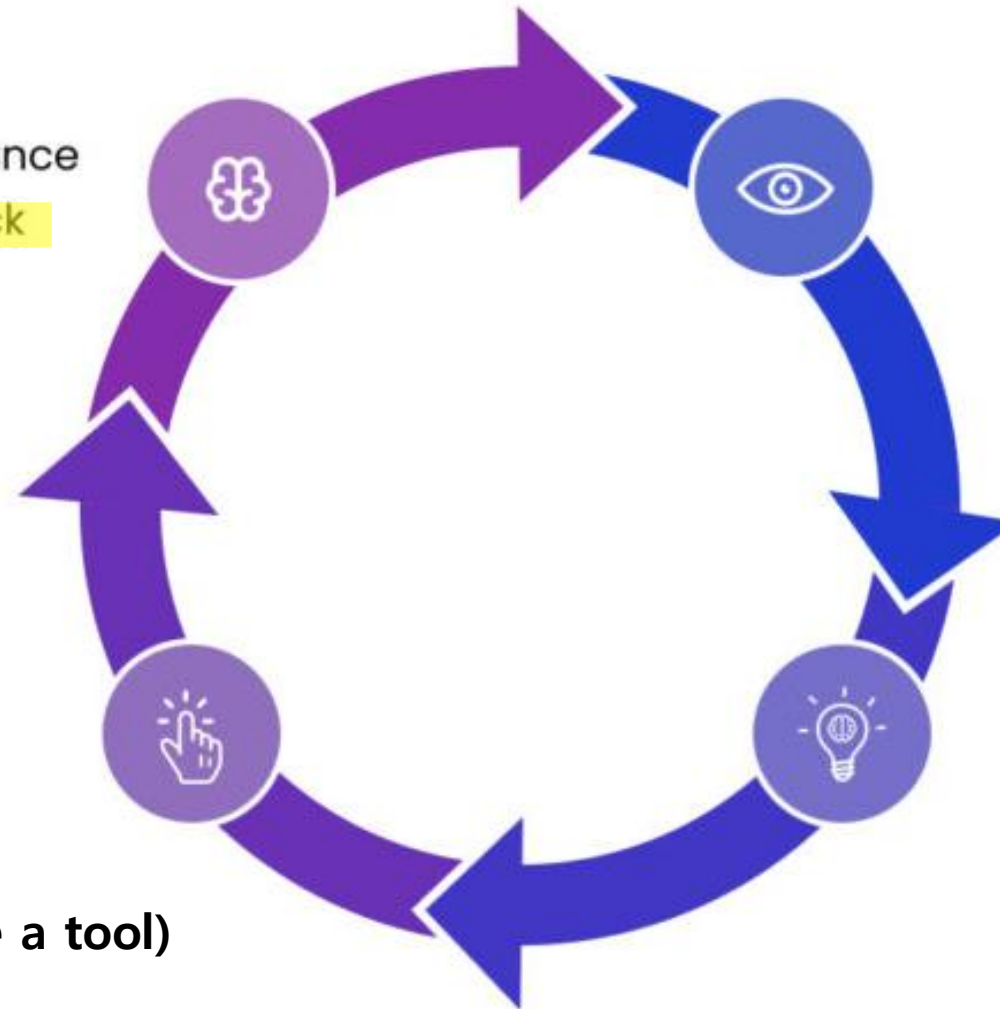
Think

Interpret data and
select action

✂ Decide what to do next

Act

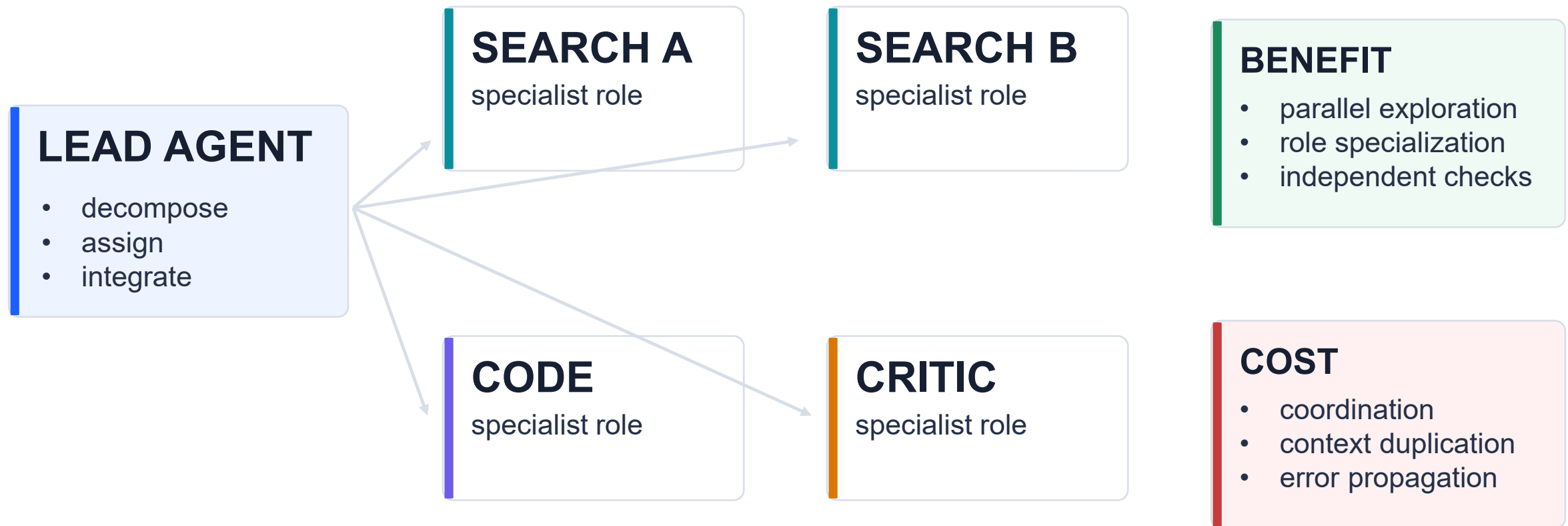
Execute action
in environment



(call an API, generate text, use a tool)

Why Multiple Agents? For complex task

Parallelism and independent checking can help — but coordination is not free.



Parallel exploration and specialization

Agent Failure Modes Are System Failure Modes

Planning drift

The agent loses the original objective over many steps.

Compounding errors

Small mistakes become irreversible downstream.

Tool misuse

Wrong command, API parameters, file, or UI target.

Cost explosion

Loops and retries consume time and tokens.

Security

Prompt injection, data exfiltration, unsafe actions.

Evaluation

Task success is harder to measure than next-token accuracy.

Production agents need permissions, sandboxes, observable state, stop conditions, and deterministic checks when possible.

05

Building Interactive Intelligence

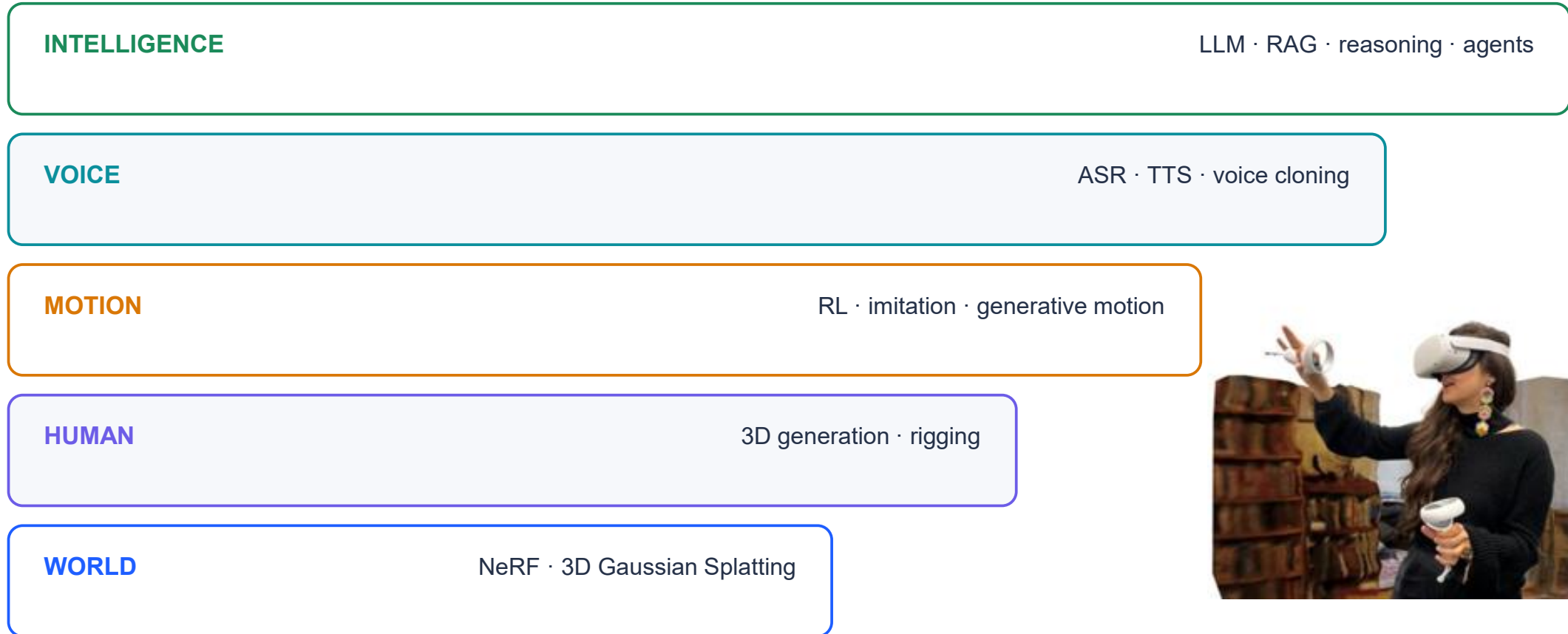
Question?

What components do we need if we want to place an intelligent AI in interactive 3D worlds?

Deep Learning Application in 2026

The Interactive Intelligence Stack

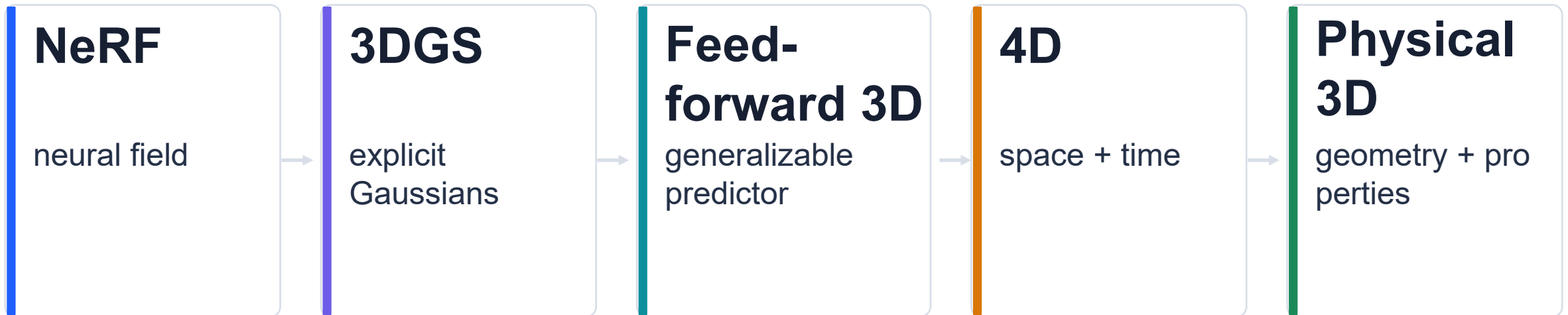
The integration — not an isolated model — is the final artifact.



TERM PROJECT · Combine multiple capabilities into one working XR interactive system.

From Pixels to Persistent 3D Worlds using SPATIAL AI

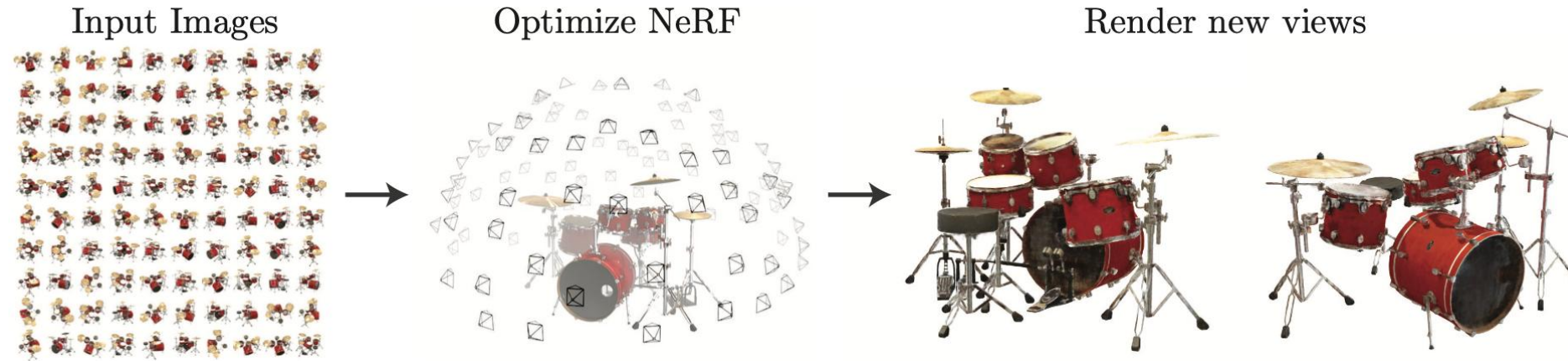
NeRF → 3DGS → feed-forward reconstruction → 4D → physical scene representations



**The trend: per-scene optimization → generalizable prediction
→ dynamic and physically useful scene representations**

NeRF: A Neural Network Represents the Scene

Week 5 question: How can a neural network generate a view it has never seen?



NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis, Mildenhall et al. 2020, ECCV

**Learn a continuous radiance field,
then volume-render novel views**



<https://arxiv.org/pdf/2003.08934>
<https://github.com/yenchenlin/nerf-pytorch>

3D Gaussian Splatting: Explicit, Fast, Differentiable

Why XR cares: explicit spatial primitives can be rendered very quickly.

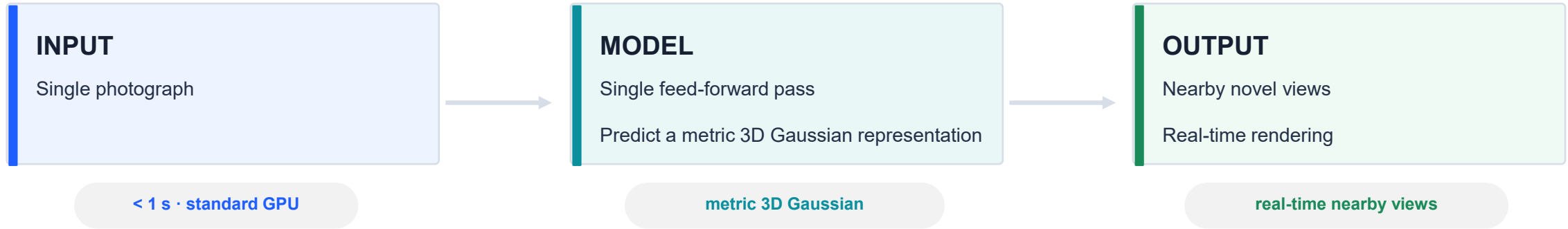


2026 direction: predict Gaussians directly; add semantics, dynamics, and physical attributes.



SOTA Snapshot · SHARP

Apple ML Research · presented at ICLR 2026



Pedagogical point: the model predicts a renderable 3D scene representation directly from a single image.



2026 Spatial AI: Reconstruction Becomes Unified

Do not memorize the acronyms — notice the direction.

❖ iLRM · CVPR 2026

- Iterative Large 3D Reconstruction Model
- Feed-forward & scalable 3D Gaussian reconstruction.

❖ UNIFIED 3D + SEMANTICS

- Generalizable scene representations increasingly combine geometry with open-vocabulary semantics.

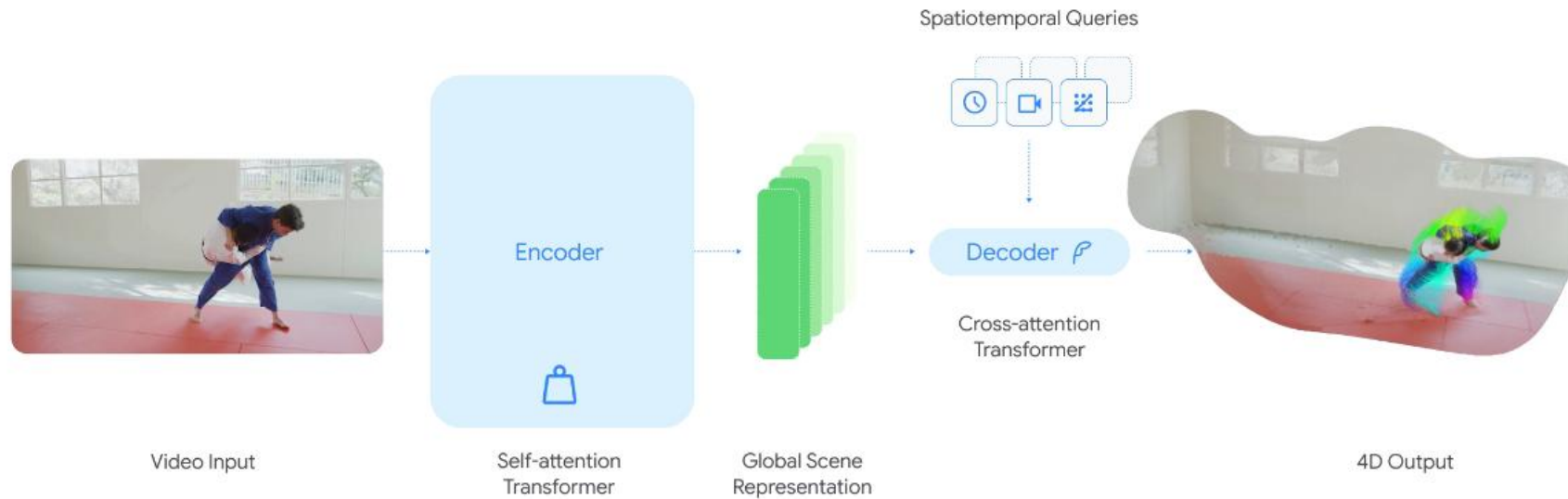
❖ D4RT · 2026

- One framework for dynamic 4D reconstruction + tracking across space and time.

Reconstruction is shifting from “make a pretty view” toward persistent perception for interactive systems.

4D Reconstruction & Tracking

Google DeepMind / D4RT: Teaching AI to see the world in four dimensions (2026)



Reconstruct geometry and track motion across space and time in one model.

<https://deepmind.google/blog/d4rt-teaching-ai-to-see-the-world-in-four-dimensions/>

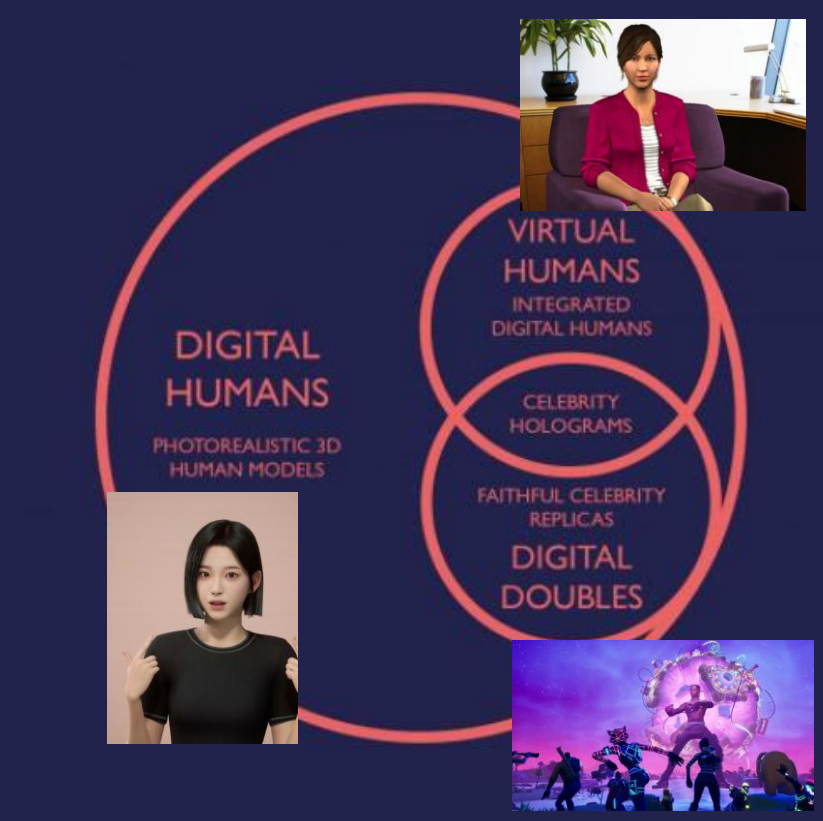
06

Digital Humans & Embodied Motion

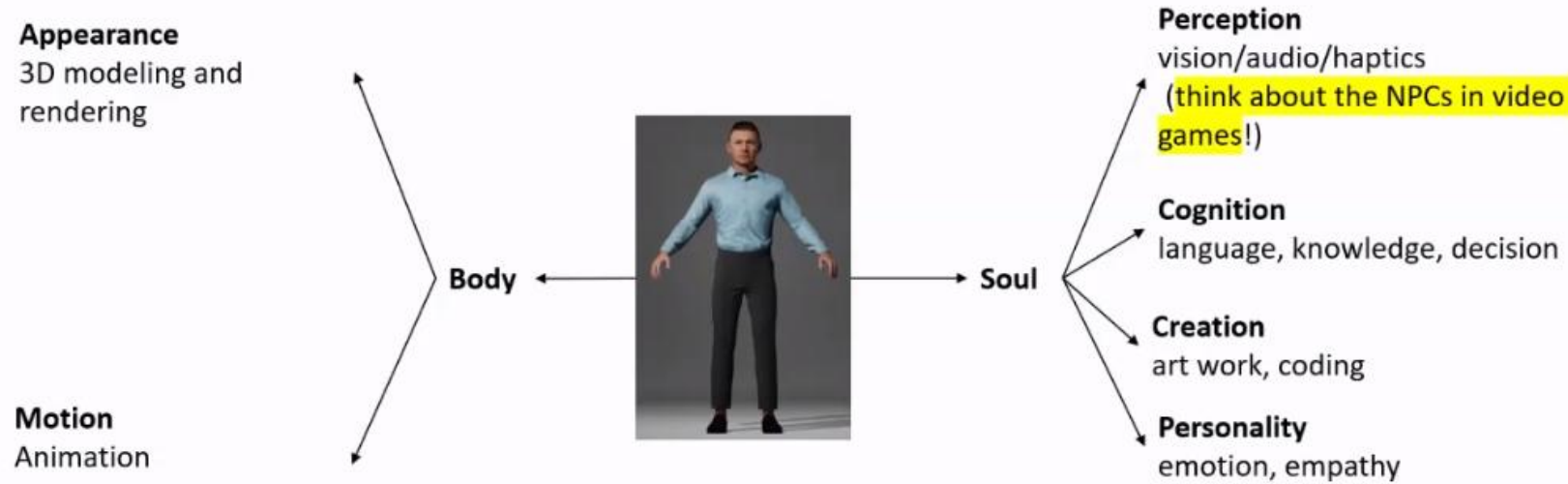
A 3D human is not just geometry — it must be rigged, animated, controlled, and eventually grounded in physics.

Deep Learning Application in 2026

What Makes a Digital Human Interactive?



Digital Human = Body + Soul



DNN & large generative models are super powerful, but we're in **the driver's seat**

2026 Direction: From Static Reconstruction to Animatable Avatars

The research target increasingly includes pose, clothing, motion-dependent appearance, and fast personalization.

INPUT

One or a few images

Possibly sparse motion / pose cues



GENERATIVE / RECONSTRUCTION PRIOR

Learned 3D human prior

Geometry + appearance + articulation



ANIMATABLE OUTPUT

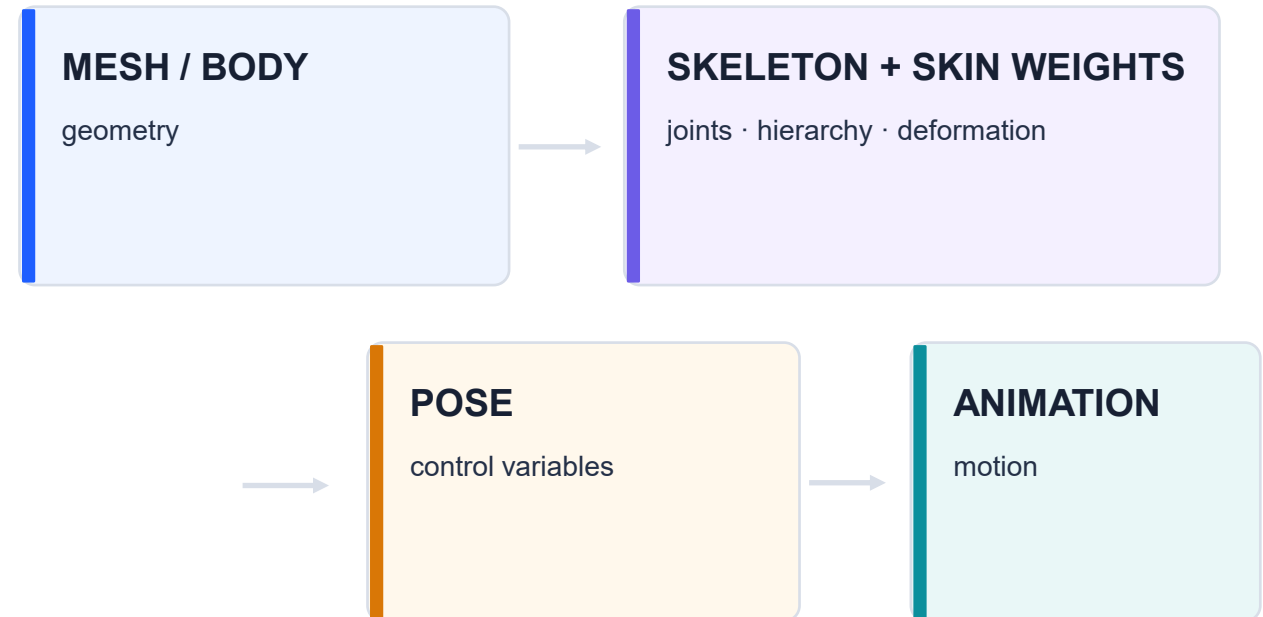
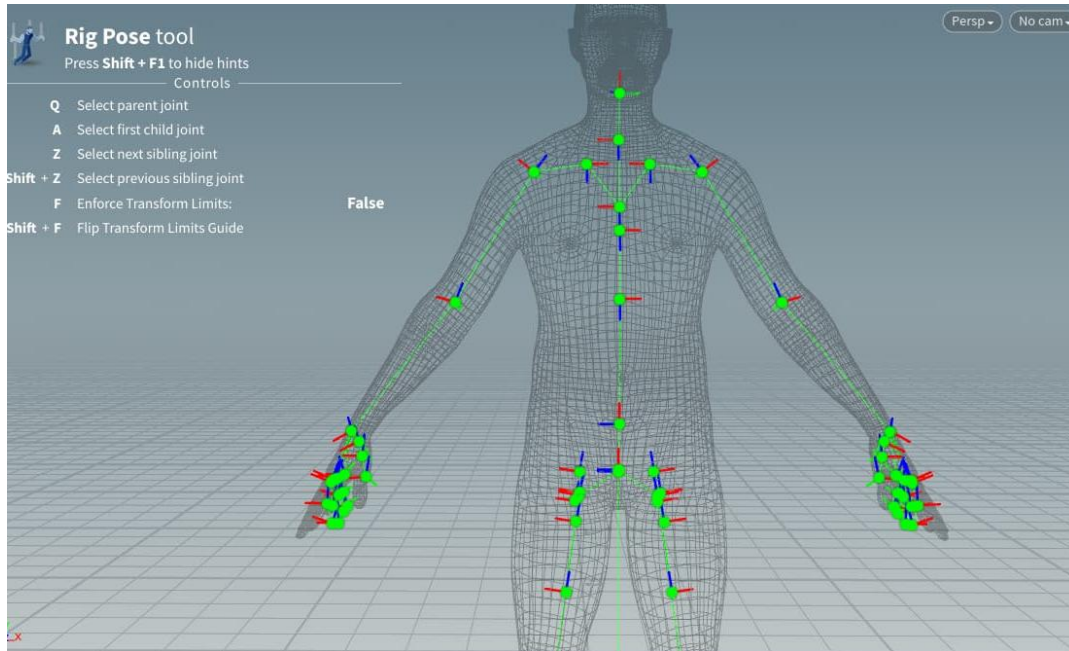
Avatar that can be posed / driven

Increasingly models cloth and motion-dependent effects

Examples in 2026 research include zero-shot animatable-human reconstruction and large-scale avatar pretraining.

Rigging Turns Geometry into a Controllable Body

Week 7 · RigNet lineage



A representation becomes useful for interaction when it exposes control variables.

This principle reappears in world models and robotics: representations matter because agents must act through them.

DeepMimic: From Playing Back Motion to Learning How to Act

Why learn an older paper? It marks the transition from motion playback to feedback control.

ANIMATION PIPELINE

Motion capture



retarget



playback

The motion is authored / recorded first.

POLICY LEARNING

Reference motion + physics simulation

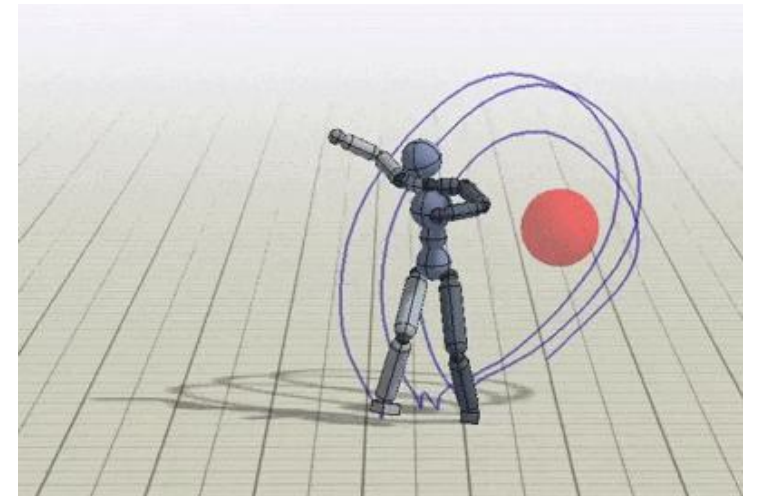
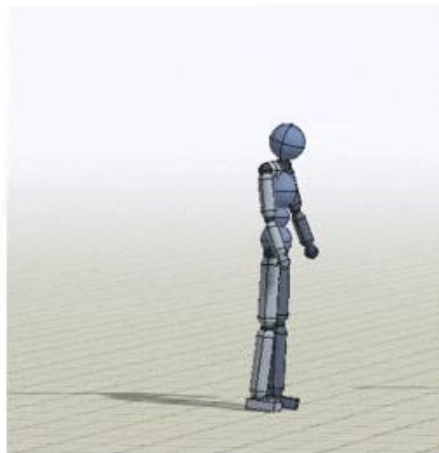
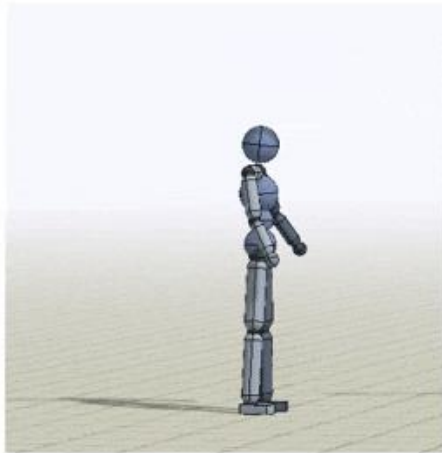
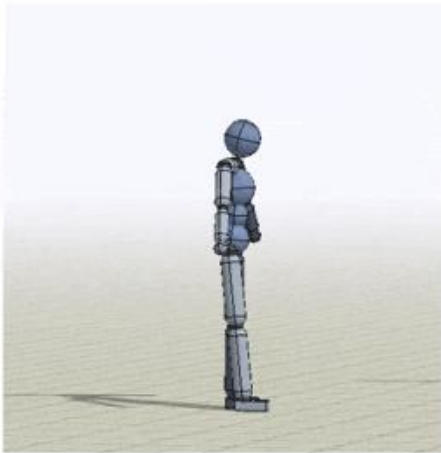


Policy $\pi(a | s)$



Physically simulated character

The controller learns actions.



2018. DeepMimic: Example-Guided Deep Reinforcement Learning of Physics-Based Character Skills, SIGGRAPH

2026 Motion Generation: Real-Time + Controllable

NVIDIA ARDY · SIGGRAPH 2026

<https://research.nvidia.com/labs/sil/projects/ardy/>

INPUT CONTROL

Online text prompts
root paths / waypoints
full-body keyframes
sparse joint constraints

ARDY

Autoregressive diffusion
streaming generation
long-horizon constraints
real-time responsiveness

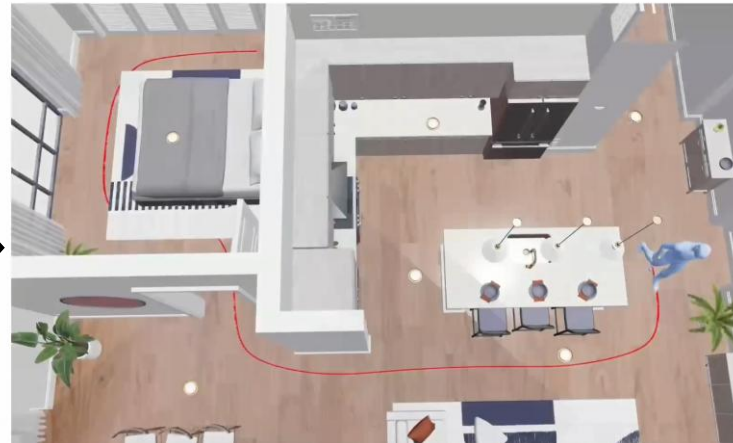
OUTPUT / USE

Interactive human motion
animation / simulation
can bridge to humanoid control with a tracking controller

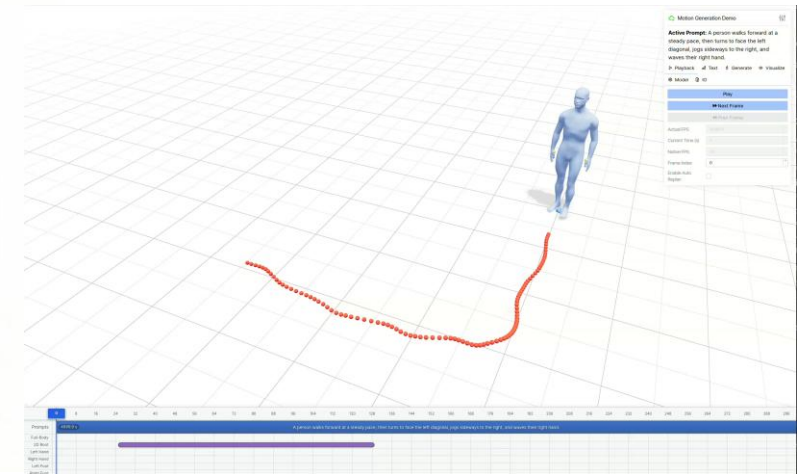


2018 Siggraph AISA

Neural State Machine for Character-Scene Interactions



Application: Scene Traversal



Research direction: combine generative motion priors with responsive control loops.

07

Speech & Voice

From ASR/TTS pipelines to native speech-to-speech interaction and controllable voices.

Deep Learning in 2026

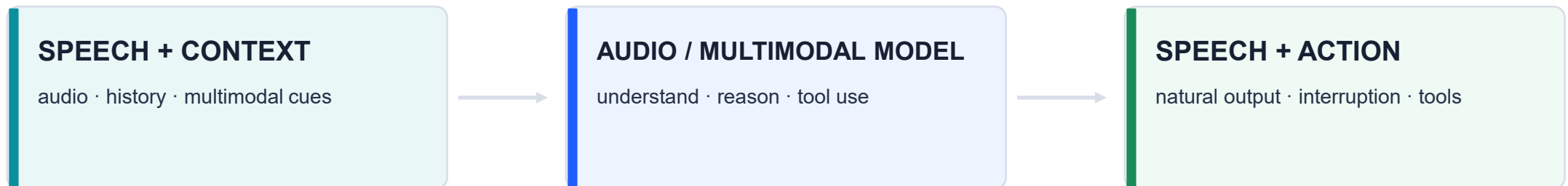
Speech AI: Pipeline → Native Conversation

Course Weeks 11–12

CLASSICAL PIPELINE



NATIVE SPEECH MODEL



Latency, prosody, turn-taking, and interruption handling become part of interaction design.

2026 SOTA Direction: Controllable Zero-Shot Speech

DISENTANGLE

- Content
- Prosody
- Timbre

Separate what is said, how it is said, and who so sounds like the speaker.

MODEL

Codec tokens + language model

Prosody continuation from a style prompt

Target timbre injected at decoding

AUTHOR-REPORTED RESULT

Competitive voice cloning

Superior zero-shot prosody control

Audio samples publicly available

**Voice synthesis is no longer only “copy a voice”;
controllability and speaker consent become system-level design issues.**

2026: Speech Models Become Interactive Agents

Voice is moving from an I/O wrapper around text toward a first-class interaction modality.

LISTEN

low-latency streaming
audio

UNDERSTAND

context + multimodal
reasoning

ACT

tools / external systems

SPEAK

natural, controllable
output

**Speech can be fused with gaze, head / hand pose,
scene context, and spatial audio inside the same agent loop.**

08

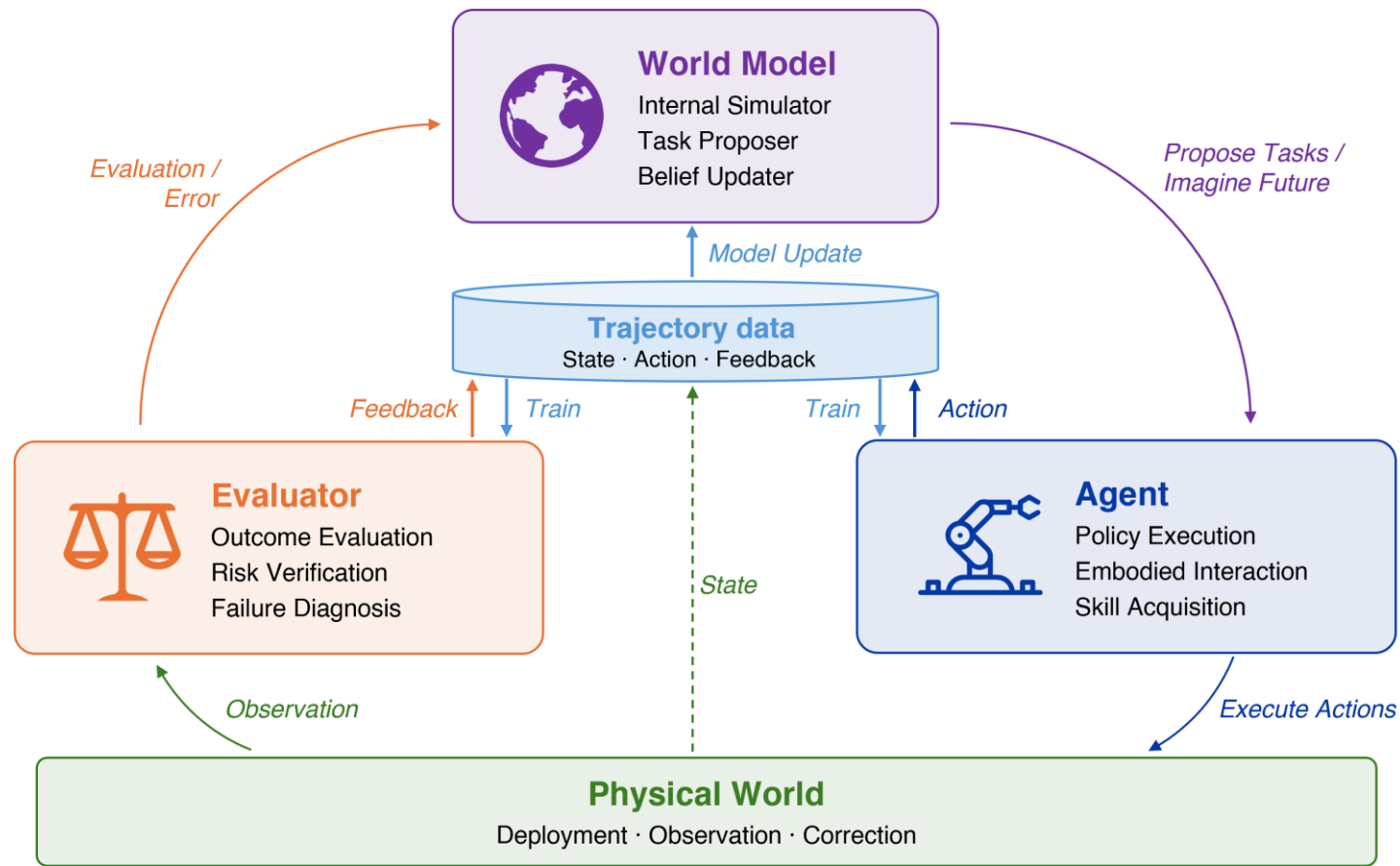
World Models & Physical AI

Generation becomes action-conditioned simulation; agents must predict what the world will do next.

Deep Learning in 2026

What Is a World Model?

A useful minimal definition: predict how the world changes when actions are taken.



PERCEIVE → PREDICT → PLAN → ACT → OBSERVE → repeat

Video Generator vs. World Model

The boundary is getting blurry; action-conditioned closed-loop behavior is the key distinction.

VIDEO GENERATOR

Input prompt / image



Generate a sequence



Watch the result

Primary question: “Does the sequence look coherent?”

WORLD MODEL

Current state + user / agent action



Generate / predict next state



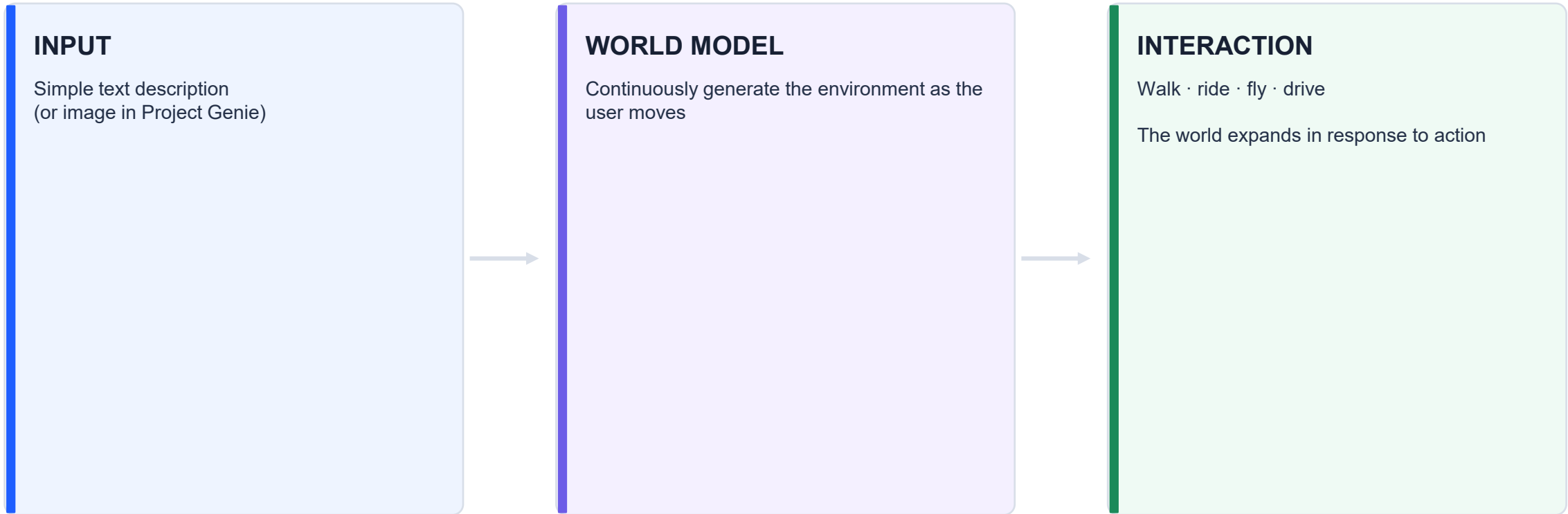
Interact and continue

Primary question: “Does the world respond coherently to action over time?”

2026 extension: NVIDIA Cosmos 3 explores omnimodal world models spanning language, image, video, audio, and actions for Physical AI.

Genie 3: Generate a World You Can Explore

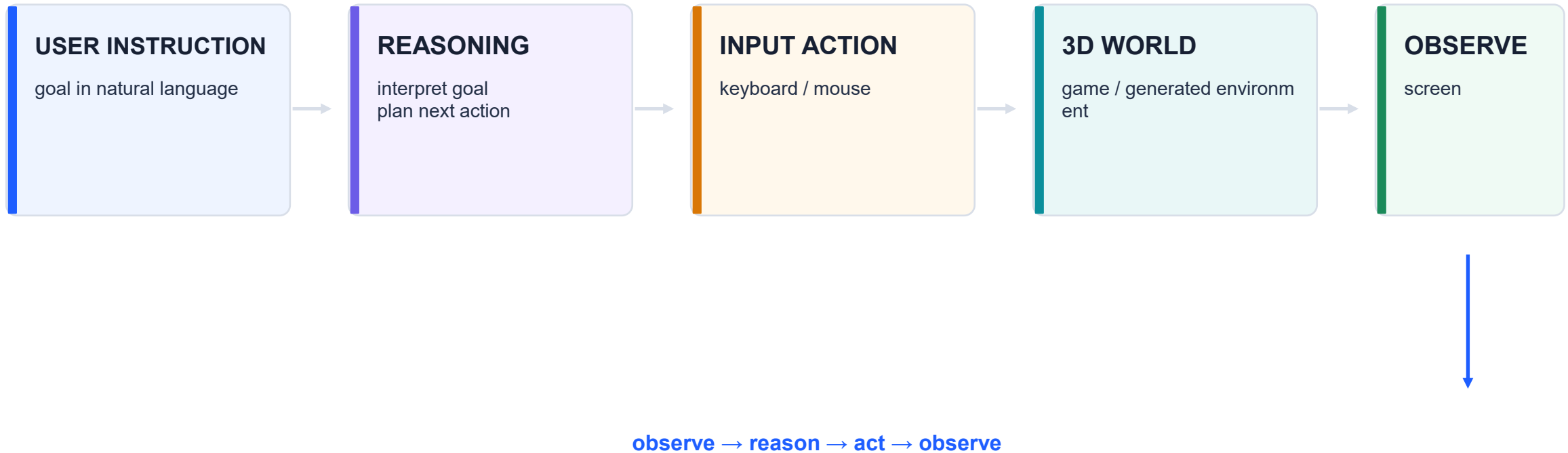
Google DeepMind · general-purpose interactive world model



A world model produces an interactive trajectory, not just a pre-rendered clip.

SIMA 2: Put an Agent Inside the World

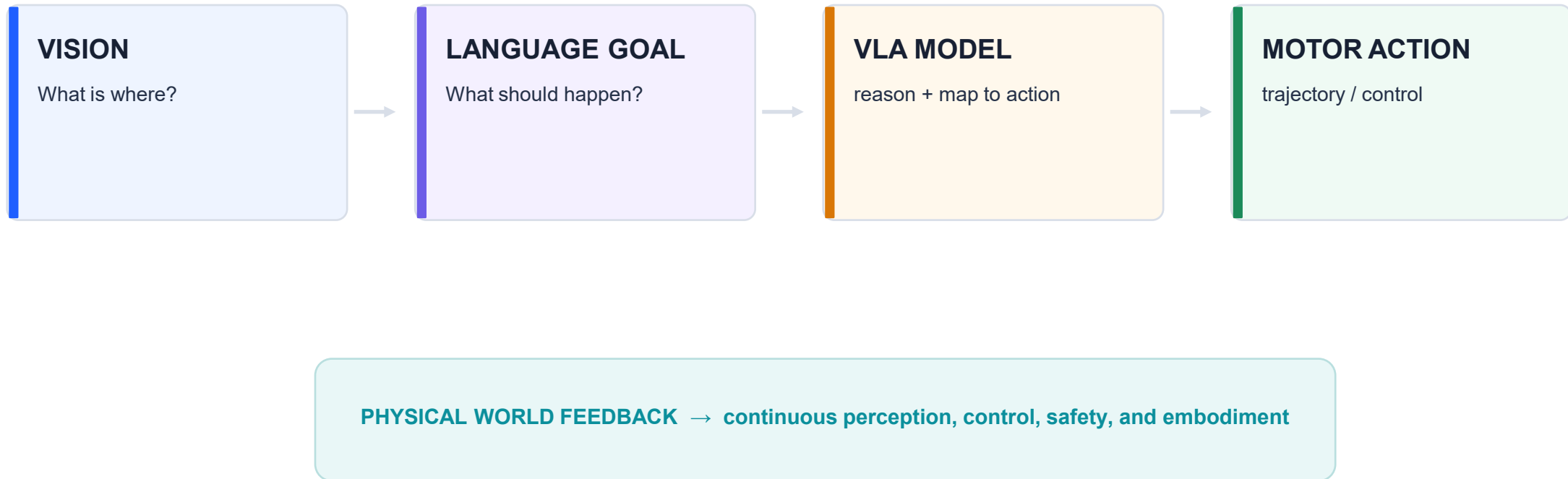
A clean bridge from agentic AI to XR-like 3D environments.



Research direction: generalist agents act through visual observations and ordinary controls, including in previously unseen 3D worlds.

Vision–Language–Action (VLA)

The “output token” can encode an action trajectory or control command — not only text.



Gemini Robotics 2 is a 2026 example of whole-body VLA control; ER 2 adds high-level embodied reasoning and orchestration.

Summary of Today

1 MULTIMODAL

AI is expanding from text and images to audio, video, 3D, and action.

2 DELIBERATIVE

Inference is becoming a process: reason, search, verify, use tools, retry.

3 INTERACTIVE

The frontier is shifting from producing answers to building systems that act in digital and physical worlds.

Model quality still matters. But in 2026, system design matters just as much.

Return to the Opening Question

Why can an AI solve an Olympiad problem
— but still struggle to clean a room?

Because

- physical world is noisy and uncertain.
- Objects follow continuous physics.
- Actions must happen in real time.
- Small errors accumulate over long sequences.
- Mistakes have physical consequences.
- And every home is different.

A digital reasoning system can often retry.



A physical agent has to survive the consequences of its actions.

- ➔ Strong performance on a digital benchmark does not automatically mean robust embodied intelligence.
- ➔ Gap between digital intelligence and embodied intelligence is the most important research frontiers.